

RPMBT XXIII – Quantum Simulation School

	9 Sept	10 Sept	11 Sept
8:30-9:00	Registration		
9:00-9:30	Welcome+Sponsors	THEORY 3 VQE, intro to errors and their mitigation	THEORY 5 ancilla-free spectral function, circuit knitting
9:30-10:30	THEORY 1 Quantum Simulations		
10:30-11:00	Break	Break	Break
11:00-12:30	EXERCISE 1 Time evolution for Ising chain	EXERCISE 3 VQE and read-out error mitigation	THEORY 6 advanced error mitigation, LCU and QSP
12:30-14:00	Lunch	Lunch	Lunch
14:00-14:20	THEORY 2 Trotter-Suzuki decomposition, linear response	LINKS Foundation	EXERCISE 5 full protocol implementation on hardware
14:20-15:30		THEORY 4 error mitigation for quantum gates	
15:30-16:00	Break	Break	Break
16:00-17:30	EXERCISE 2 Correlation functions and execution on emulators	EXERCISE 4 Execution on quantum hardware with full mitigation	EXERCISE 6 full protocol implementation on hardware, extensions

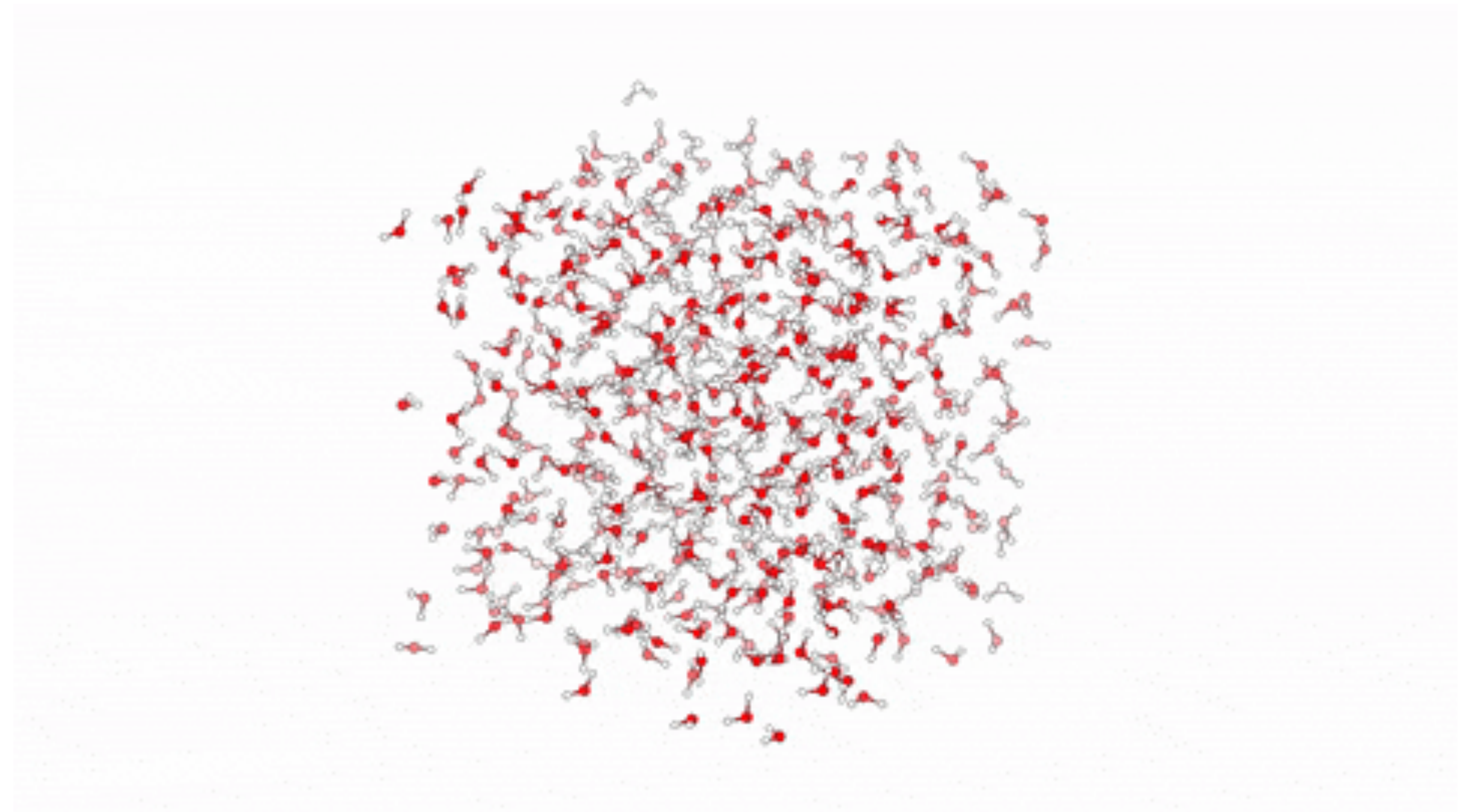
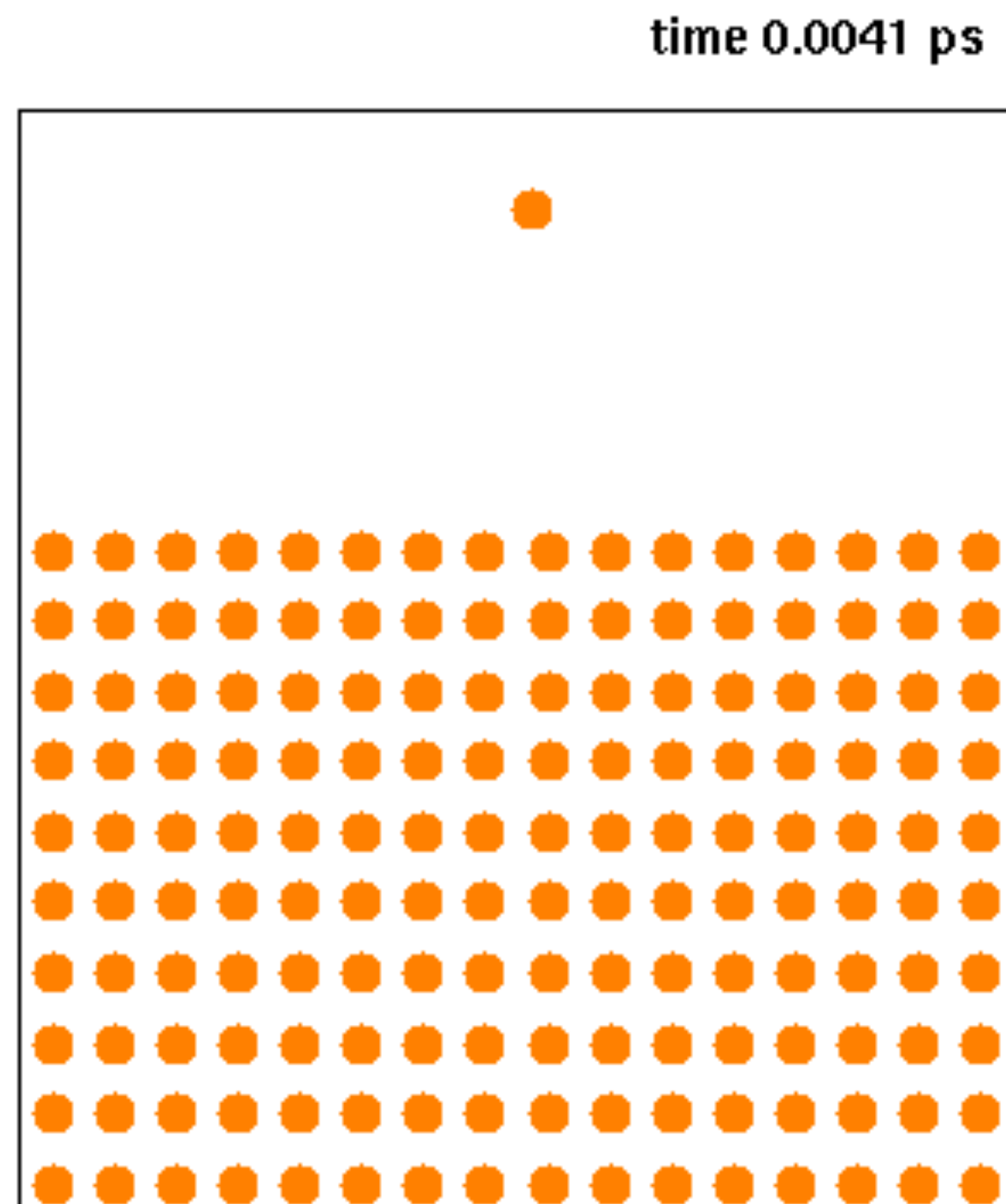
RPMBT XXIII — Quantum Simulation School

- General notion
- The Ising Hamiltonian
- Applications of time evolution: quench experiments
- Measurement and statistical uncertainties
- Observations on
 - preparation (mapping to qubits)
 - evolution (quantum circuits and controllable approximations)
 - measurement (statistical uncertainties)

Simulation

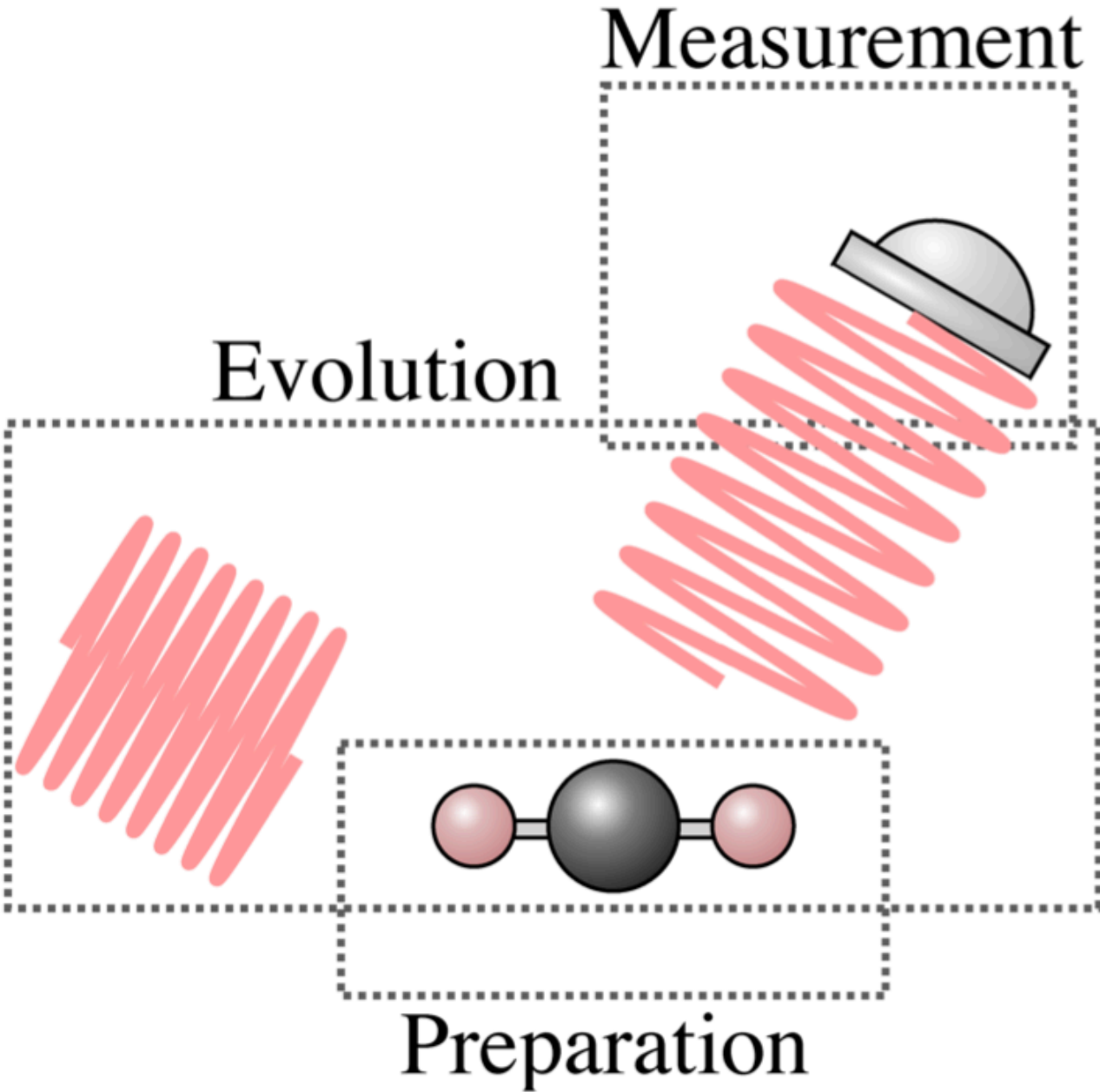
Running of a mathematical model on a computer, the model being designed to represent the behavior of, or the outcome of, a real-world physical system.

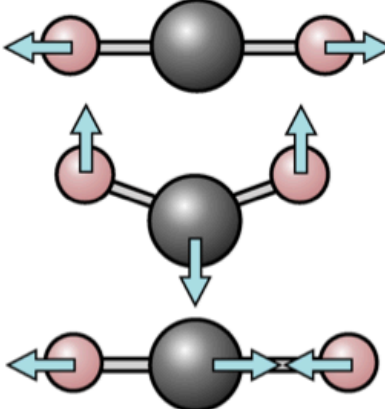
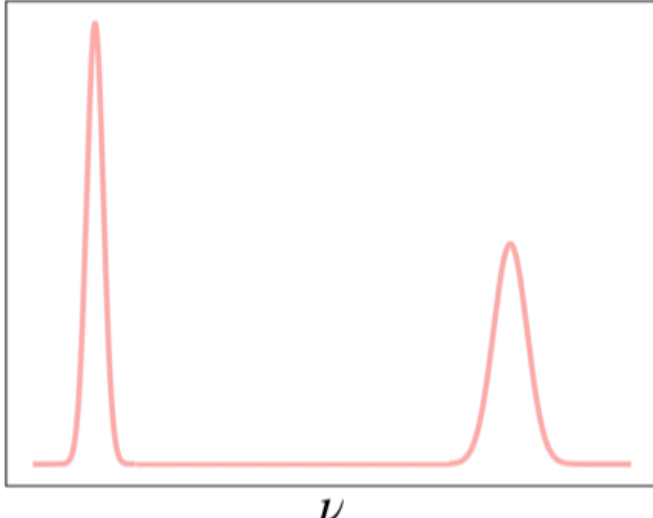
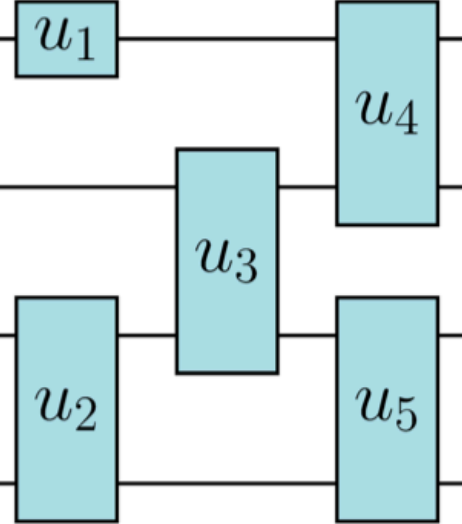
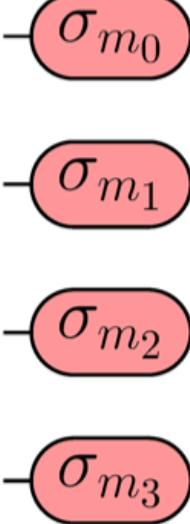
Example: molecular dynamics (Newton equation $F = ma$ with a parametrized force field).



Quantum simulation

Use a controllable quantum system (a quantum computer) to study the properties (energies, response...) of another less controllable/accessible quantum system.

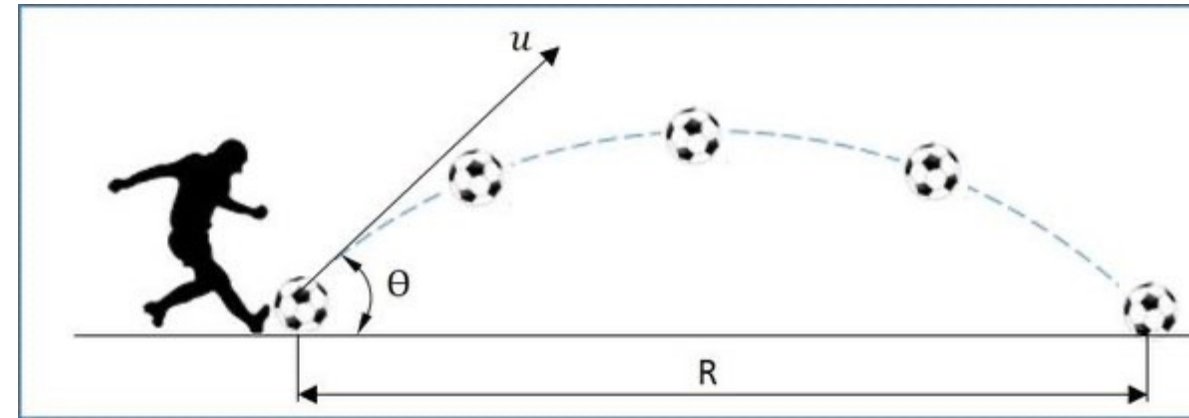


	Preparation	Evolution	Measurement
Experiment			
Quantum theory	$ \Psi\rangle$	$e^{-it\hat{H}} \Psi\rangle$	$\langle\Psi \hat{B} \Psi\rangle$
Quantum computation	$ 0\rangle -$ $ 0\rangle -$ $ 0\rangle -$ $ 0\rangle -$		

Simulation of time evolution

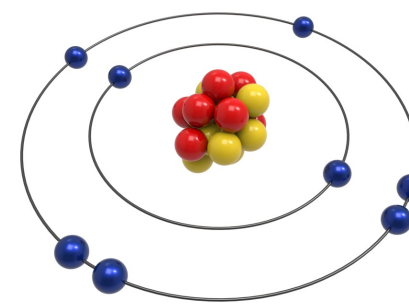
Classical Mechanics: Newton's equation

$$\vec{F} = m\vec{a}$$
$$\vec{x}(0), \vec{v}(0) \rightarrow \vec{x}(t), \vec{v}(t)$$



Quantum Mechanics: Schrodinger's equation

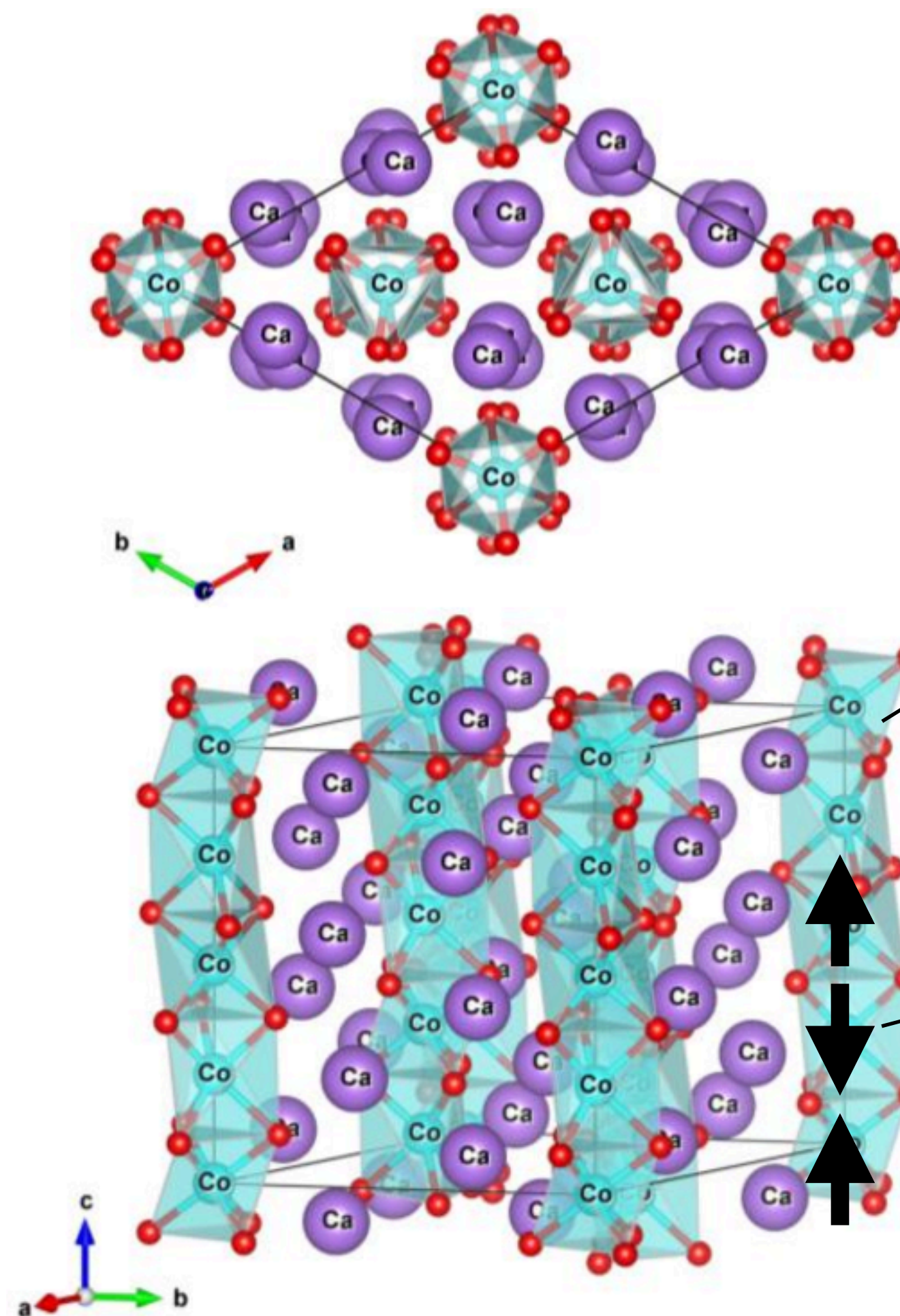
$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$
$$|\Psi(0)\rangle \rightarrow |\Psi(t)\rangle$$



Effective spin Hamiltonians for magnetic materials

The magnetic properties of certain materials can be understood modeling them as systems of spins.

Example: in $\text{Ca}_3\text{Co}_2\text{O}_6$ and CoNb_2O_6 the Co^{2+} ions have a magnetic momentum or “spin”



spin pointing up $\uparrow$ maps to $|0\rangle$

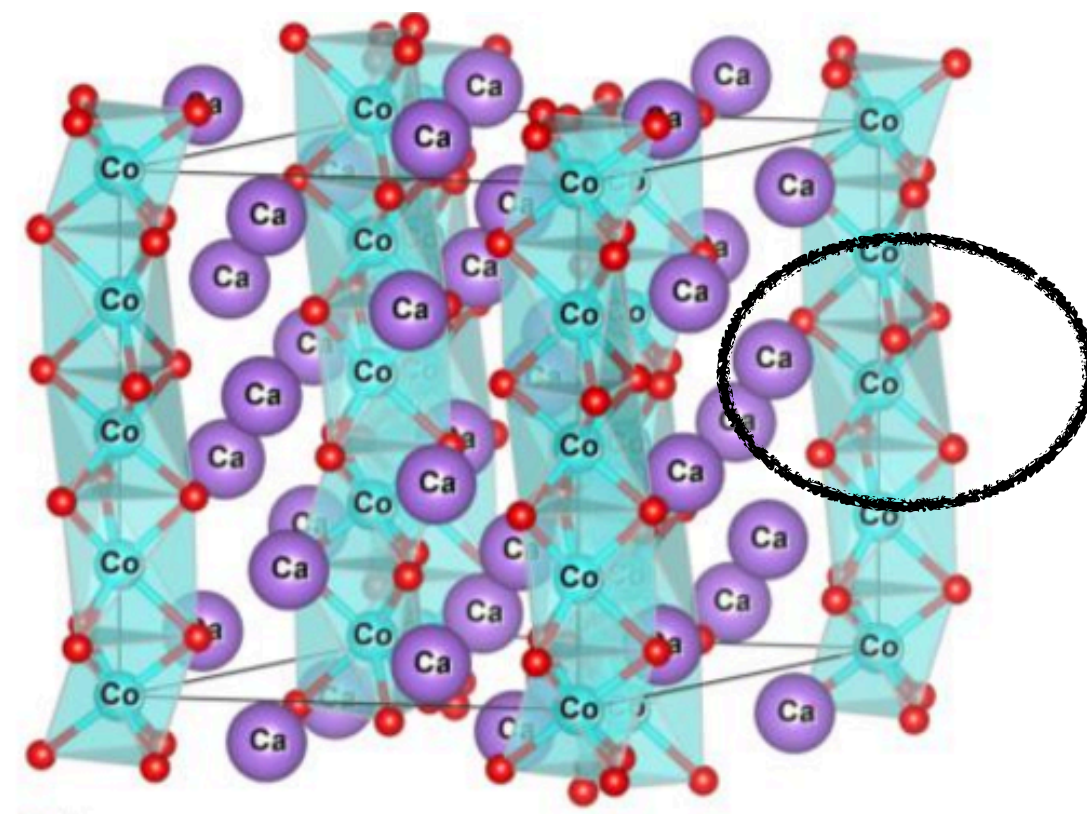
spin pointing down $\downarrow$ maps to $|1\rangle$

state $\uparrow\downarrow\uparrow$ maps to $|010\rangle$

<https://www.nature.com/articles/s41467-019-13273-4>

<https://www.science.org/doi/10.1126/science.1180085>

The Hamiltonian — case of a single spin



spin pointing up $\uparrow$ maps to $|0\rangle$

spin pointing down $\downarrow$ maps to $|1\rangle$

If a magnetic field is applied to the particle, its energy is described by the operator

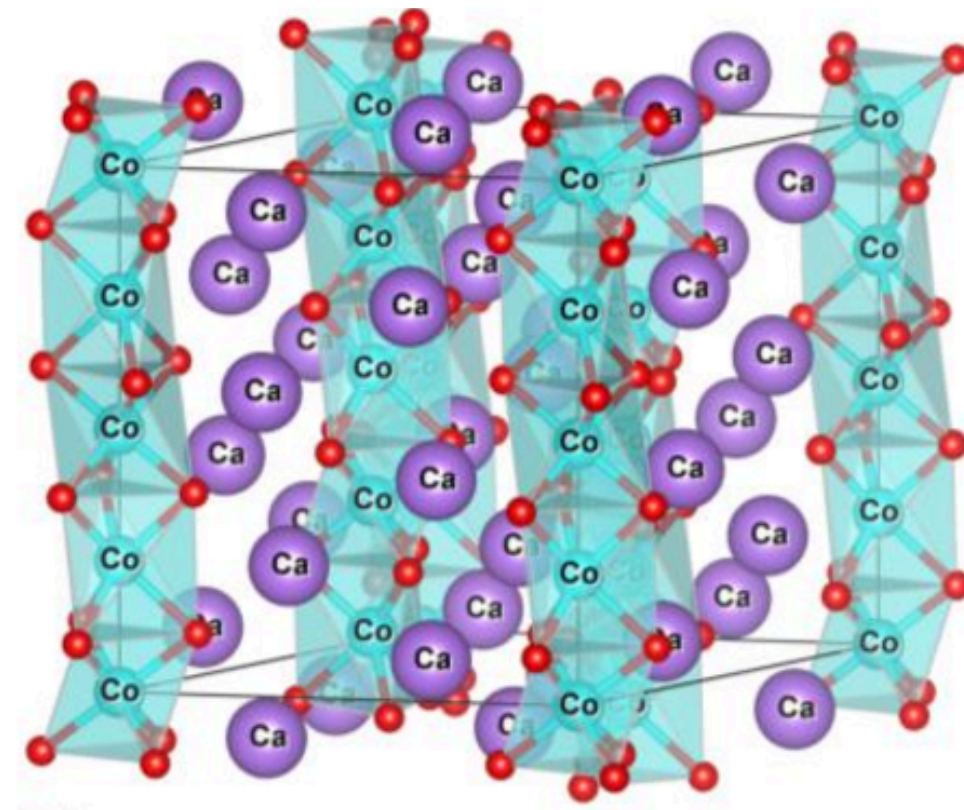
$$\hat{H} = -\hat{\mu} \cdot \mathbf{B} = -\mu B \left[n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = -\mu B [n_x X + n_y Y + n_z Z]$$

If the magnetic field is in the Z direction,

$$\hat{H} = -\mu B Z = -hZ$$

The states $|0\rangle$ and $|1\rangle$ have well-defined energy, respectively $-h$ and h .

The Hamiltonian – case of multiple spins



spin pointing up $\uparrow$ maps to $|0\rangle$

spin pointing down $\downarrow$ maps to $|1\rangle$

Spins interact with each other, in a way that depends on the material under study.

The Ising interaction happens along a preferential direction (anisotropic interaction), e.g.

$$\hat{H} = -JX_1X_2$$

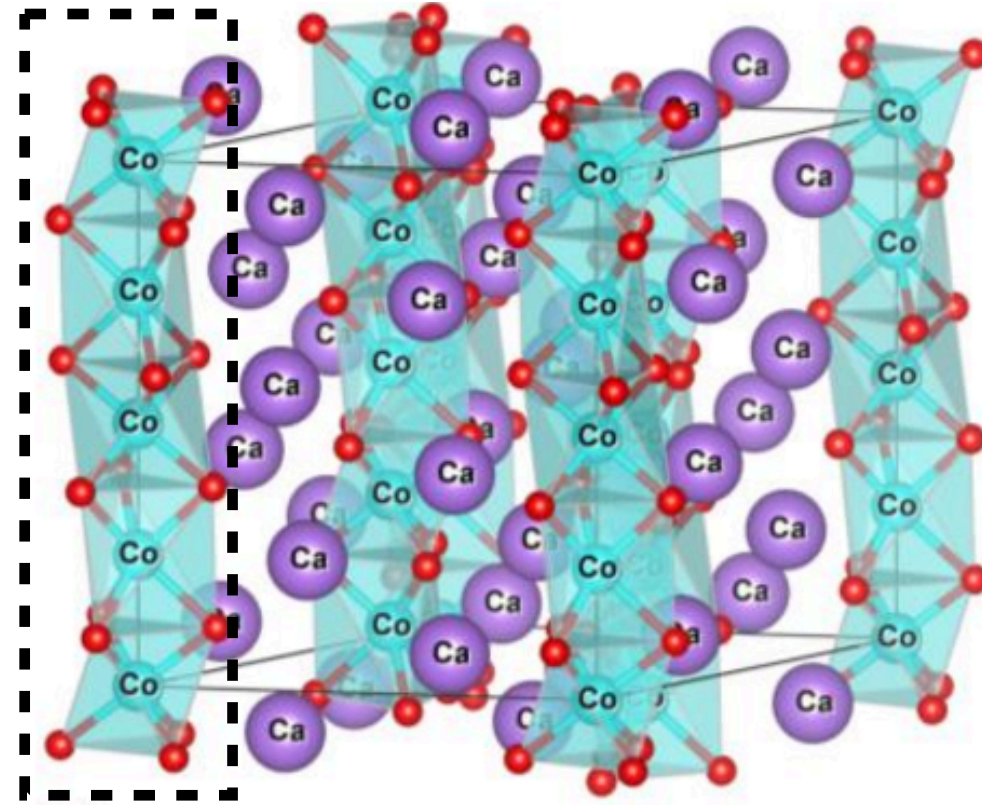
The states $\rightarrow\rightarrow$ and $\leftarrow\leftarrow$ have well-defined energy $-J$.

The states $\rightarrow\leftarrow$ and $\leftarrow\rightarrow$ have well-defined energy J .

The interaction may favor parallel alignment (ferromagnetism, if $J > 0$)
or antiparallel alignment (anti ferromagnetism, if $J < 0$).

The Ising Hamiltonian

In $\text{Ca}_3\text{Co}_2\text{O}_6$ and CoNb_2O_6 the Co^{2+} ions have a magnetic momentum or “spin”



spin pointing up $\uparrow$ maps to $|0\rangle$

spin pointing down $\downarrow$ maps to $|1\rangle$

The spins are arranged in a 1D chain.

They interact with each other along a direction (e.g. X)

They are subject to an external field along a different direction (e.g. Z)

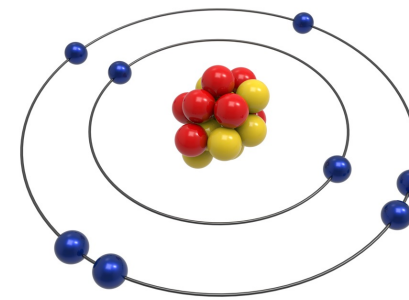
$$H = -J \sum_q X_q X_{q+1} - h \sum_q Z_q$$

Quench experiments

Quantum Mechanics: Schrodinger's equation

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

$$|\Psi(0)\rangle \rightarrow |\Psi(t)\rangle$$



1. Prepare a system in a controlled initial state applying a strong magnetic field, e.g.

$$|\psi(0)\rangle = |000\rangle$$

2. Let the system evolve under H (external field + interaction) to produce $|\psi(t)\rangle = e^{-itH} |\psi(0)\rangle$

What to measure?

Quench experiments

In a quench experiment, one may measure

- correlation functions $C_{qr}(t) = \langle \psi(t) | Z_q Z_r | \psi(t) \rangle - \langle \psi(t) | Z_q | \psi(t) \rangle \langle \psi(t) | Z_r | \psi(t) \rangle$ – to probe spread of correlations captured by the Lieb-Robinson bound $C_{qr}(t) \propto e^{-\frac{d(q,r)}{ct}}$ [1]

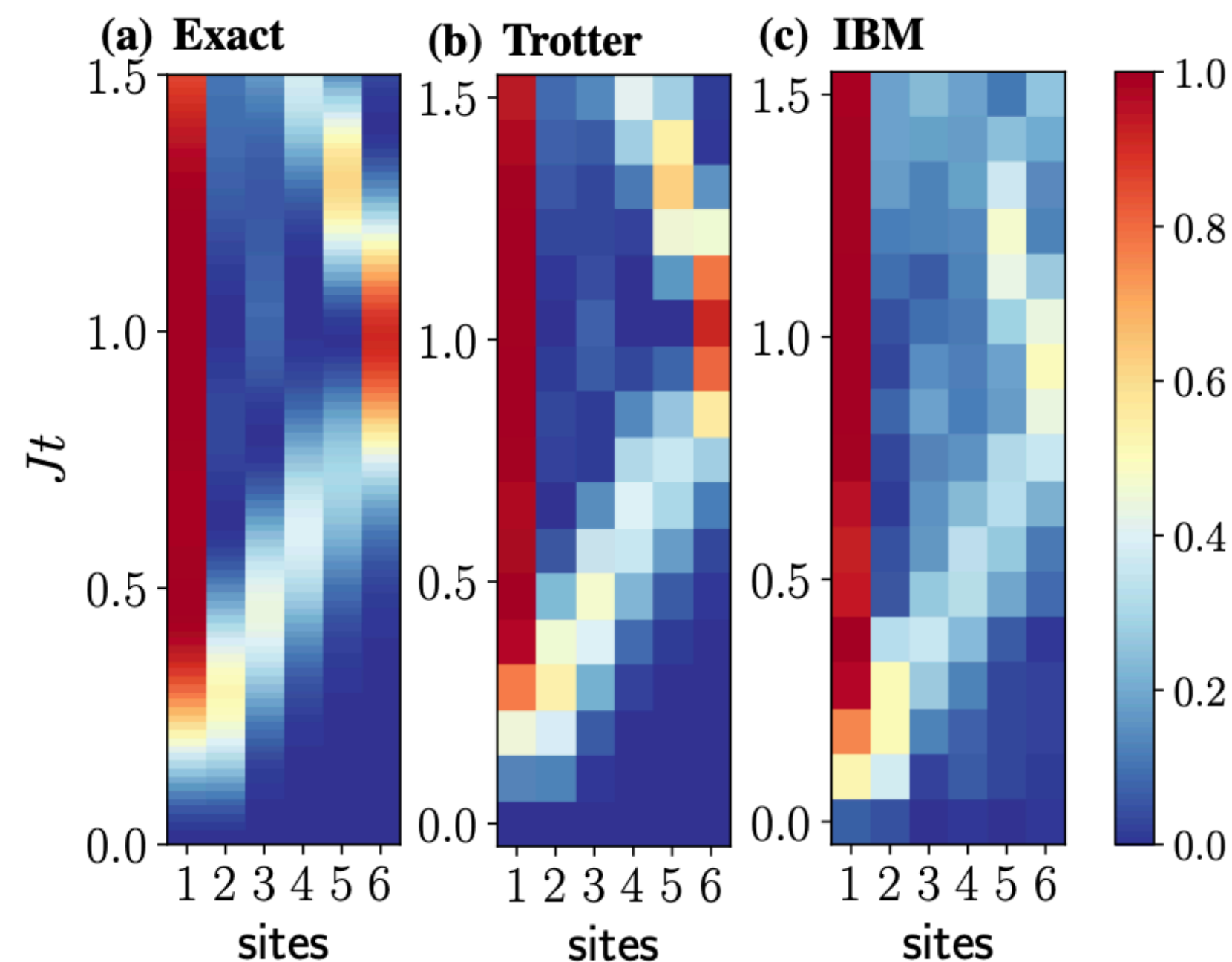


Figure 4. Results for the connected spin correlator defined in Eq. (6). Data is computed using: (a) exact diagonalization, (b) numerical trotterization, (c) the IBM quantum device. Data is shown for $N = 6, h_j = 0$ and $U = 0$, using a symmetric trotterization, and obtained on 12 March 2019.

Quench experiments

In a quench experiment, one may measure

- time-dependent expectation values $Z_q(t) = \langle \psi(t) | Z_q | \psi(t) \rangle$ to probe thermalization, i.e.

$$\lim_{t \rightarrow \infty} Z_q(t) = \text{Tr}[\rho Z_q] \text{ or memory of the initial state [1]}$$

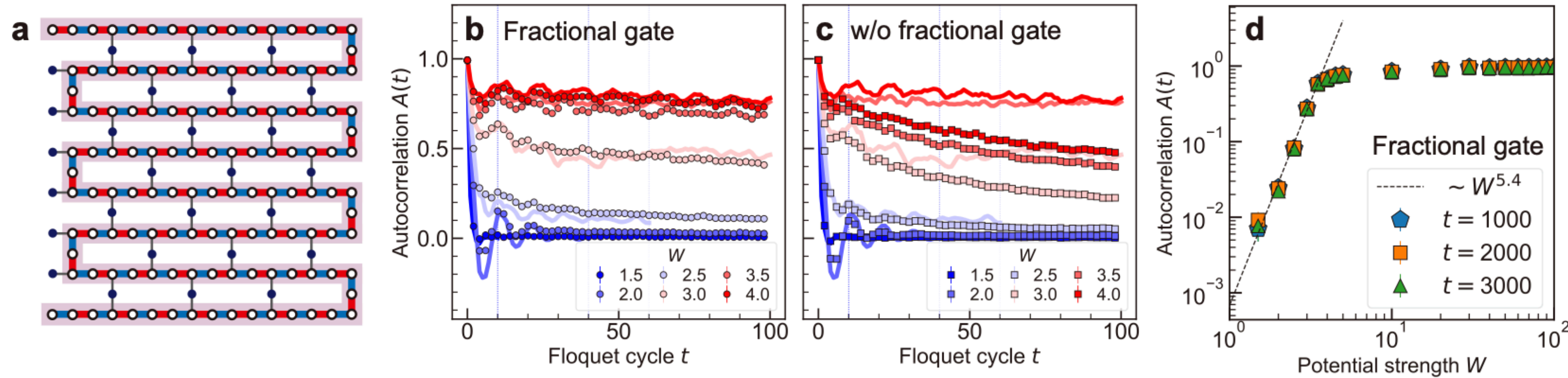


FIG. 2. **Autocorrelation function in the one-dimensional quasiperiodic (QP) kicked Ising model.** **a** Connectivity layout of qubits on the `ibm_kobe` processor. White circles highlight the 129 qubits selected for our experiments, forming a one-dimensional chain. Colored bonds represent nearest-neighbour RZZ couplings. Red (blue) bonds indicate the even (odd) layer of parallelized RZZ gates (see also Fig. 1a). **b** Experimental results obtained using fractional gates up to 100 Floquet cycles. Symbol colors indicate the strength of W . Error bars are estimated from the average measurement errors of $\hat{Z}_j$, but are sufficiently small to be negligible. For comparison, tDMRG results are shown as solid lines with the corresponding colors, up to the Floquet cycle t for which convergence is confirmed with bond dimension $\chi = 512$ (indicated by vertical dotted lines). **c** Same as **b** but without fractional gates. **d** Longer-time evolution of the autocorrelation up to 3000 Floquet cycles using fractional gates. The dashed line shows a power-law fit to the data with $W < 4.0$.

Quench experiments

In a quench experiment, one may measure

- the Loschmidt echo $R(t) = |\langle \psi(0) | \psi(t) \rangle|^2$ to probe dynamical phase transitions, i.e. non-differentiability in $L(t) = \lim_{n \rightarrow \infty} (1/n) \log R(t)$ [1]

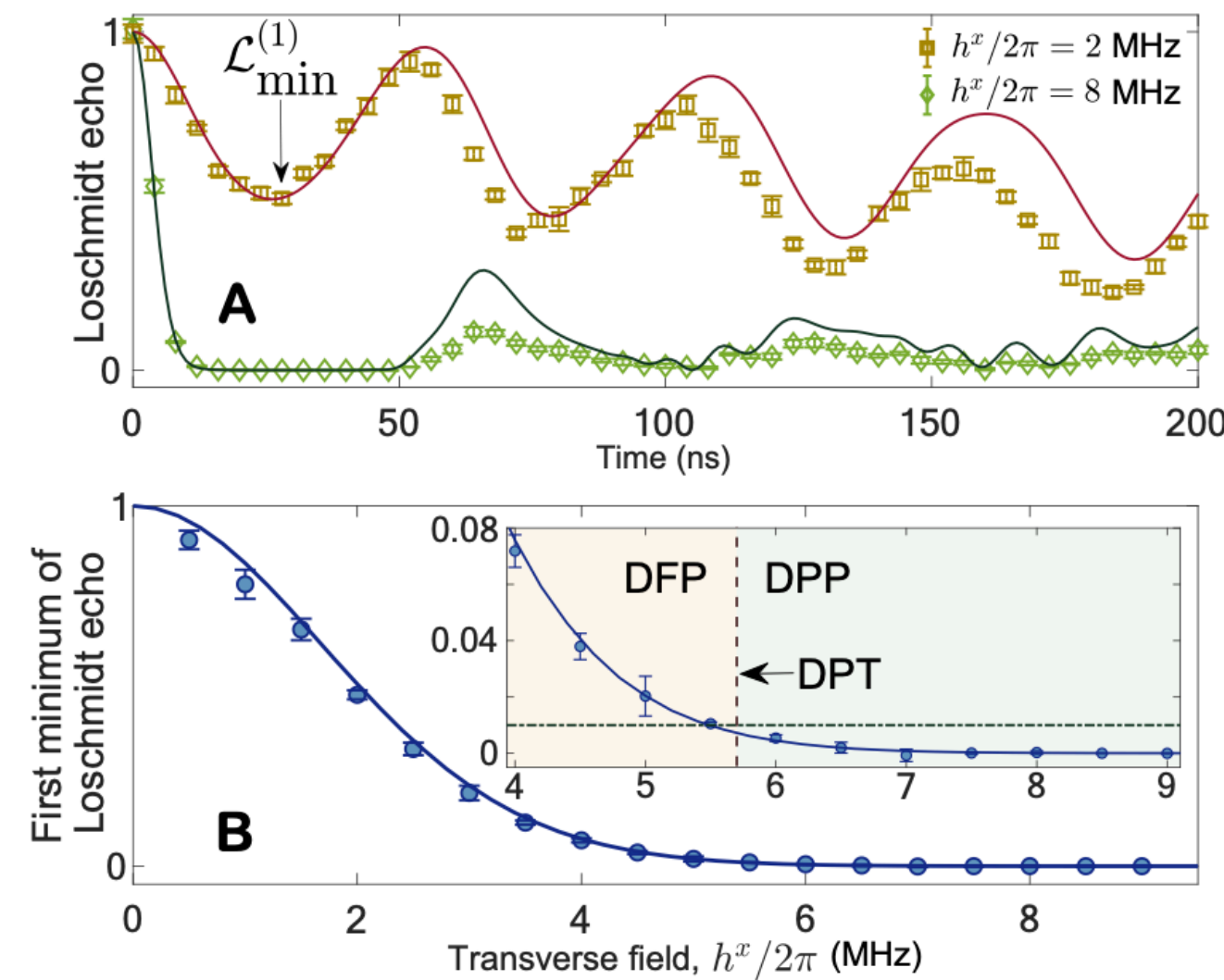
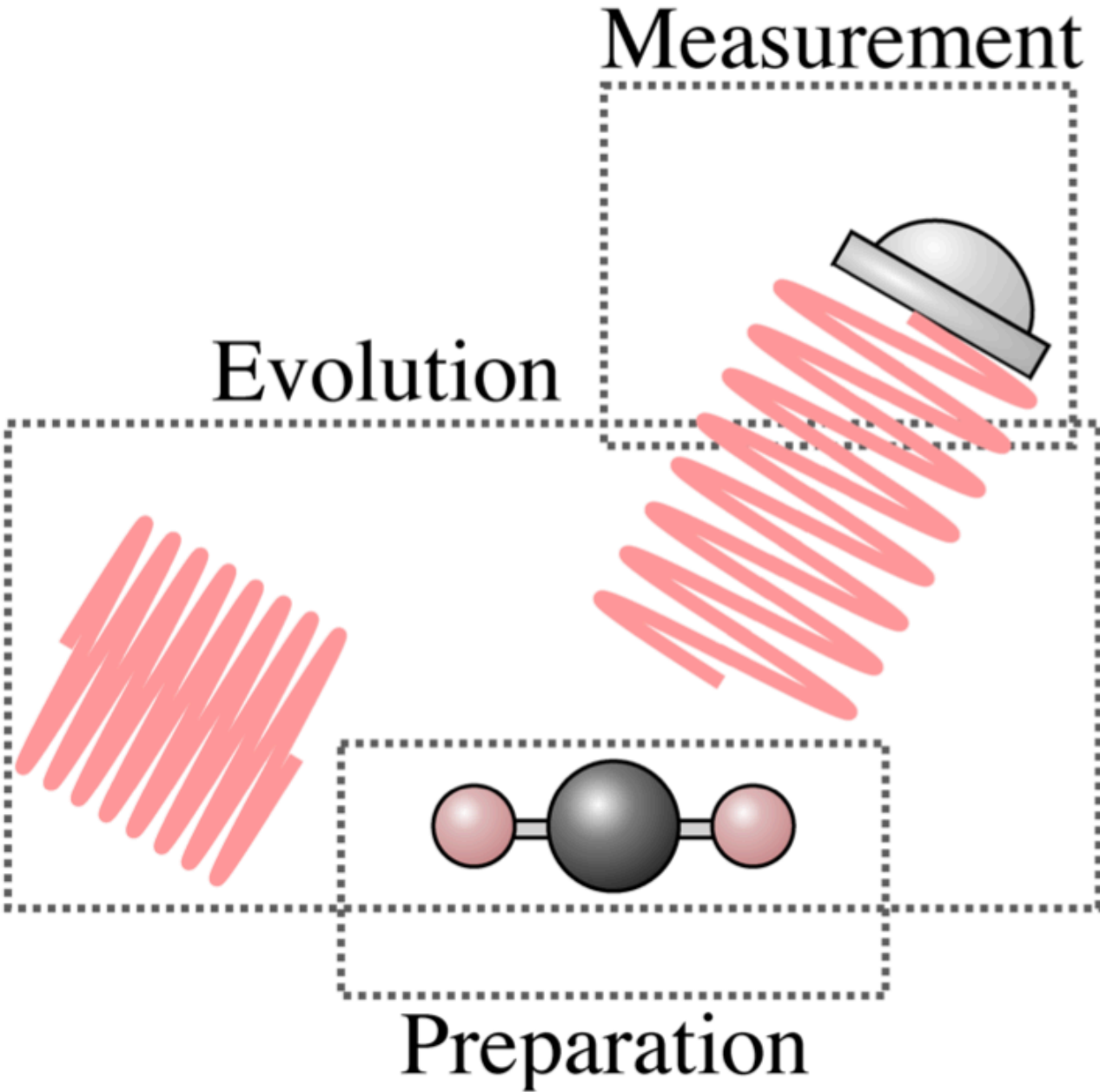
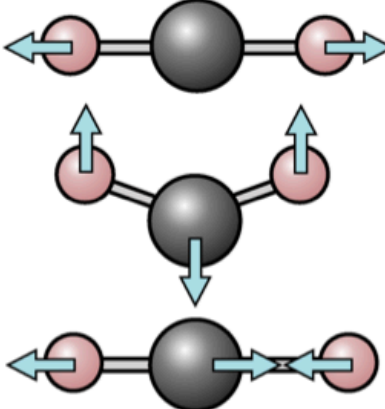
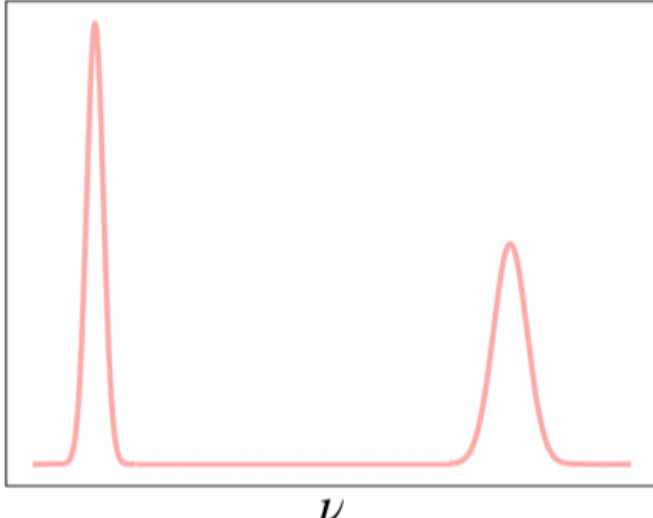
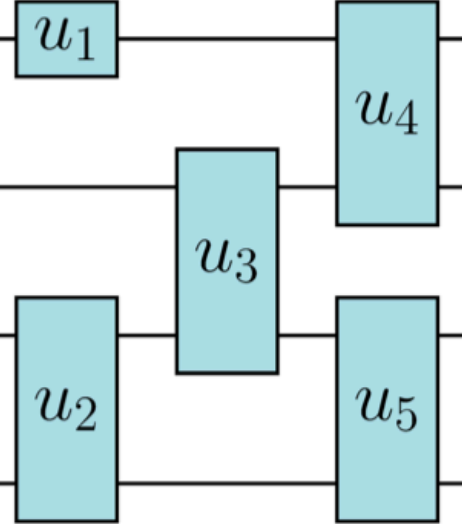
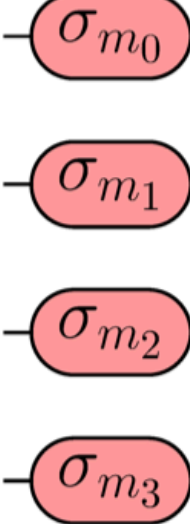


FIG. 3. **Loschmidt echo.** **A**, Time evolution of the Loschmidt echo $\mathcal{L}(t)$ for different transverse field strengths. **B**, The earliest minimum point of $\mathcal{L}(t)$ during its dynamics, $\mathcal{L}_{\min}^{(1)}$, as a function of h^x . The behaviour of $\mathcal{L}(t)$ for longer time indicates the existence of anomalous dynamical phases¹² (see Methods). Additionally, we note that the Loschmidt echo cannot be strictly equal to 0, because of finite-size effects. We demonstrate that $\mathcal{L}_{\min}^{(1)}$ reaches $\sim 10^{-2}$ at the transition point and becomes smaller in the paramagnetic phase for the LMG model with $N = 16$ (see Methods). The solid curves in **A** and **B** are the numerical results using Eq. (1) without considering decoherence.

Quantum simulation

Use a controllable quantum system (a quantum computer) to study the properties (energies, response...) of another less controllable/accessible quantum system.



	Preparation	Evolution	Measurement
Experiment			
Quantum theory	$ \Psi\rangle$	$e^{-it\hat{H}} \Psi\rangle$	$\langle\Psi \hat{B} \Psi\rangle$
Quantum computation	$ 0\rangle -$ $ 0\rangle -$ $ 0\rangle -$ $ 0\rangle -$		

Mapping system state onto qubit state

The first step to simulate Hamiltonian dynamics on a quantum computer is to map degrees of freedom (i.e., the states and operators) of the target system on those of a register of qubits.

For spin-1/2 particles, the mapping is simple and natural:

- For a single spin,

$$J|\uparrow\rangle = |0\rangle \quad , \quad J|\downarrow\rangle = |1\rangle$$

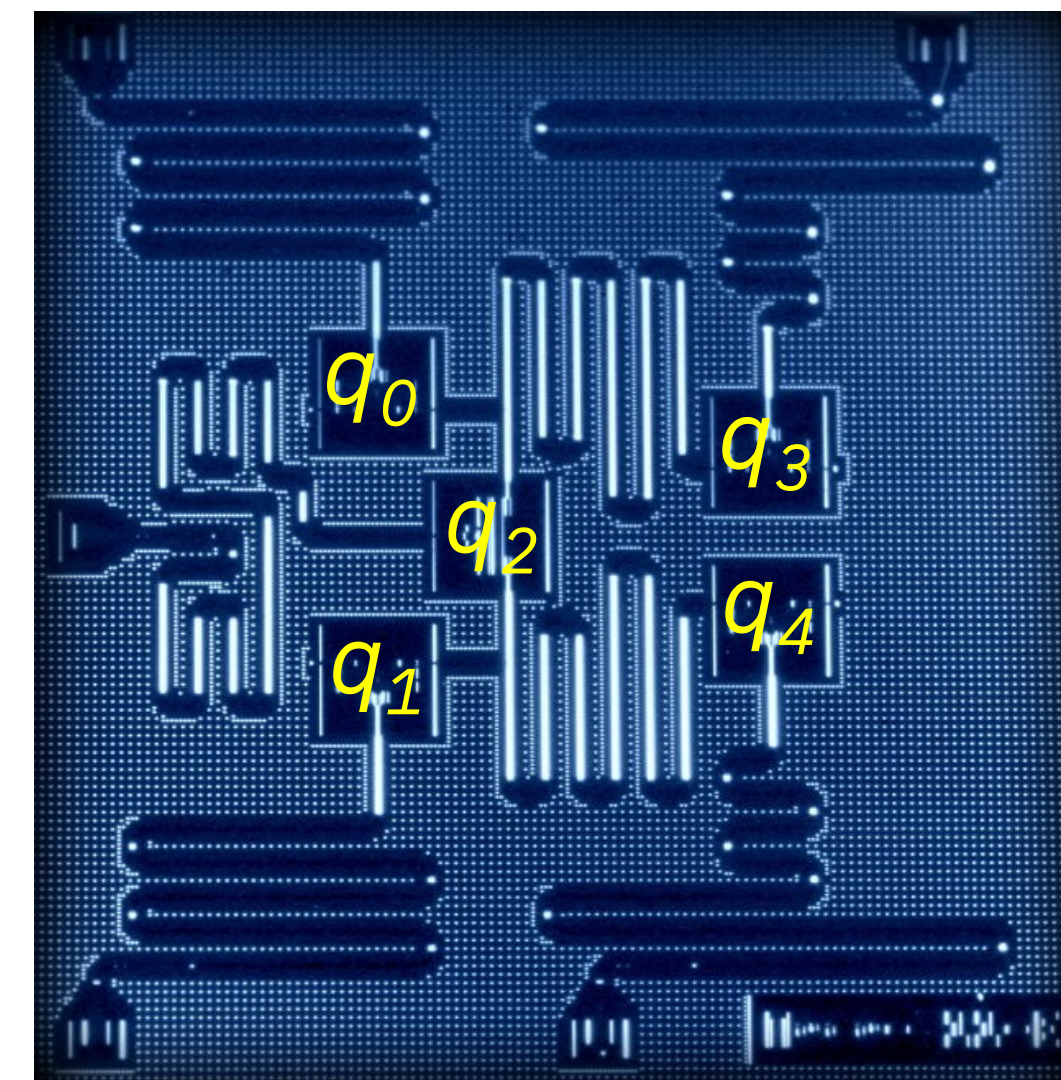
- For n spins, use n qubits and map the states of the kth spin onto those of the kth qubit

$$J|\uparrow\downarrow\downarrow\uparrow\uparrow\rangle \rightarrow |01100\rangle$$

(Qiskit: 4th, 3rd, 2nd, 1st, 0th i.e. first qubit to the right)

- Mapping states of n spins on states of n qubits defines an isomorphism J between the Hilbert spaces of spins and qubits. Operators are mapped as

$$B_{qubit} = JB_{spin}J^{-1}$$



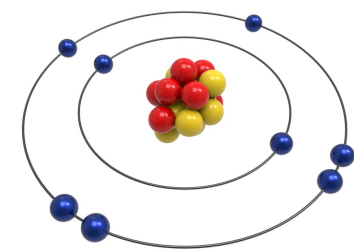
Simulating dynamics with quantum circuits

Time evolution is a unitary operator, and any quantum circuit implements a unitary operations

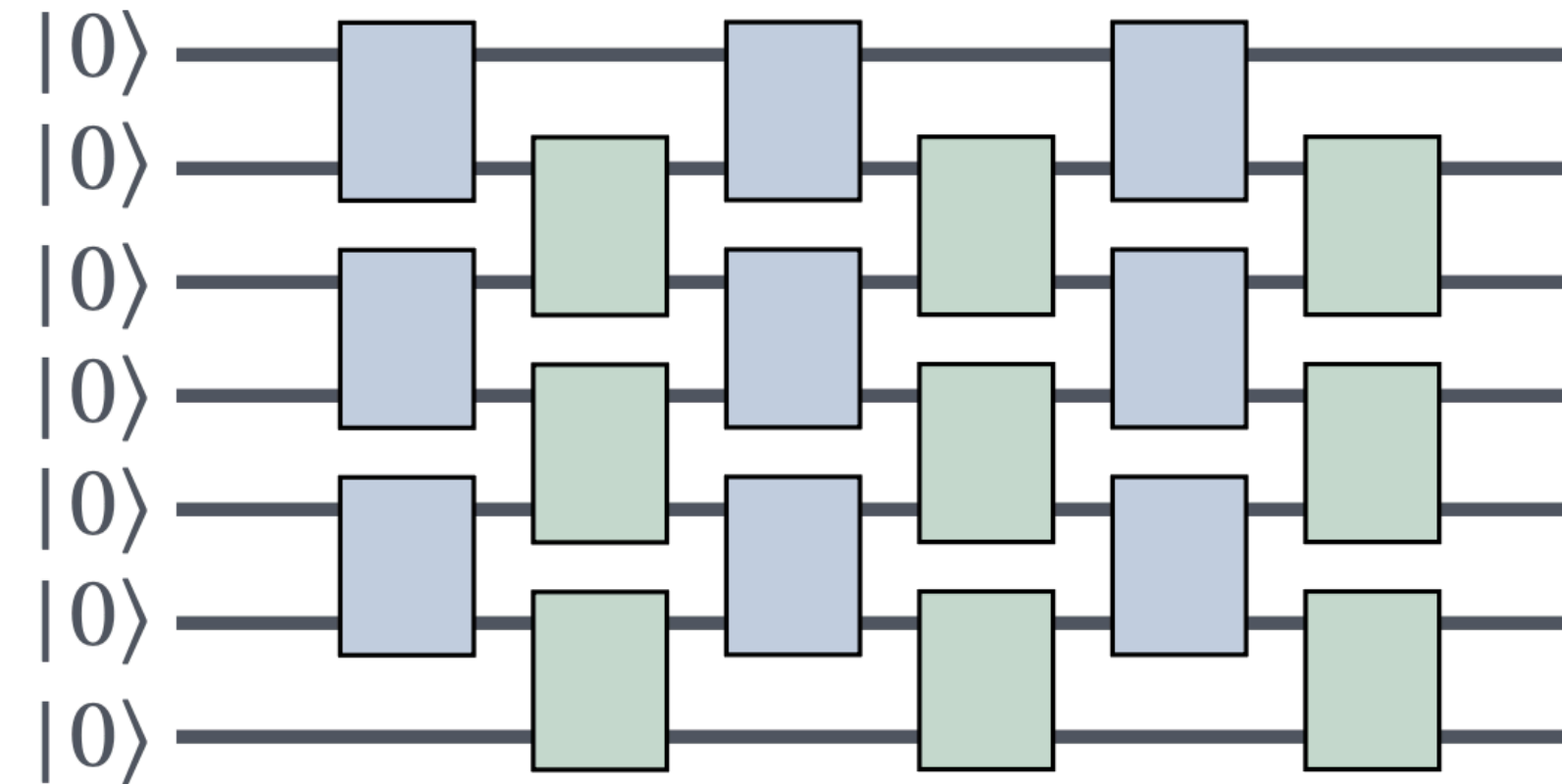
Quantum Mechanics: Schrodinger's equation

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

$$|\Psi(0)\rangle \rightarrow |\Psi(t)\rangle = e^{\frac{t\hat{H}}{i\hbar}} |\Psi(0)\rangle$$



$$|\Psi(t)\rangle = e^{\frac{tH}{i\hbar}} |\Psi(0)\rangle$$



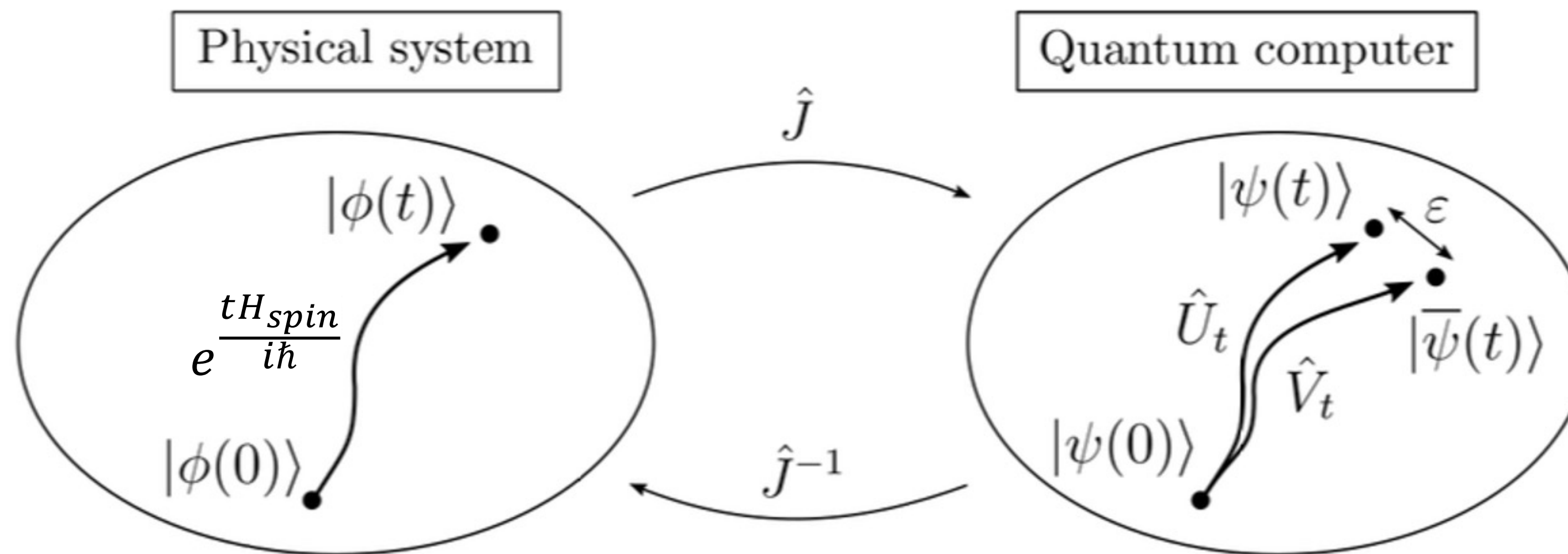
$$|\bar{\Psi}(t)\rangle = \hat{V}_t |\Psi(0)\rangle$$

To **efficiently simulate time evolution** means to produce a state that

- accurately approximates the time-dependent state, $|\Psi(t) - \bar{\Psi}(t)| < \varepsilon$
- has cost polynomial in system size n , target precision $1/\varepsilon$, and evolution time t

Simulating dynamics with quantum circuits

Mapping states of n spins on states of n qubits defines an isomorphism J between the Hilbert spaces of spins and qubits. Operators are mapped as $B_{qubit} = JB_{spin}J^{-1}$



Under the isomorphism, the time-evolution operator takes the form $\hat{U}_t = e^{\frac{tH_{qubit}}{i\hbar}} = e^{\frac{tJH_{qubit}J^{-1}}{i\hbar}} = J e^{\frac{tH_{spin}}{i\hbar}} J^{-1}$

For generic Hamiltonians, the exact exponential maps to a circuit of exponential cost.

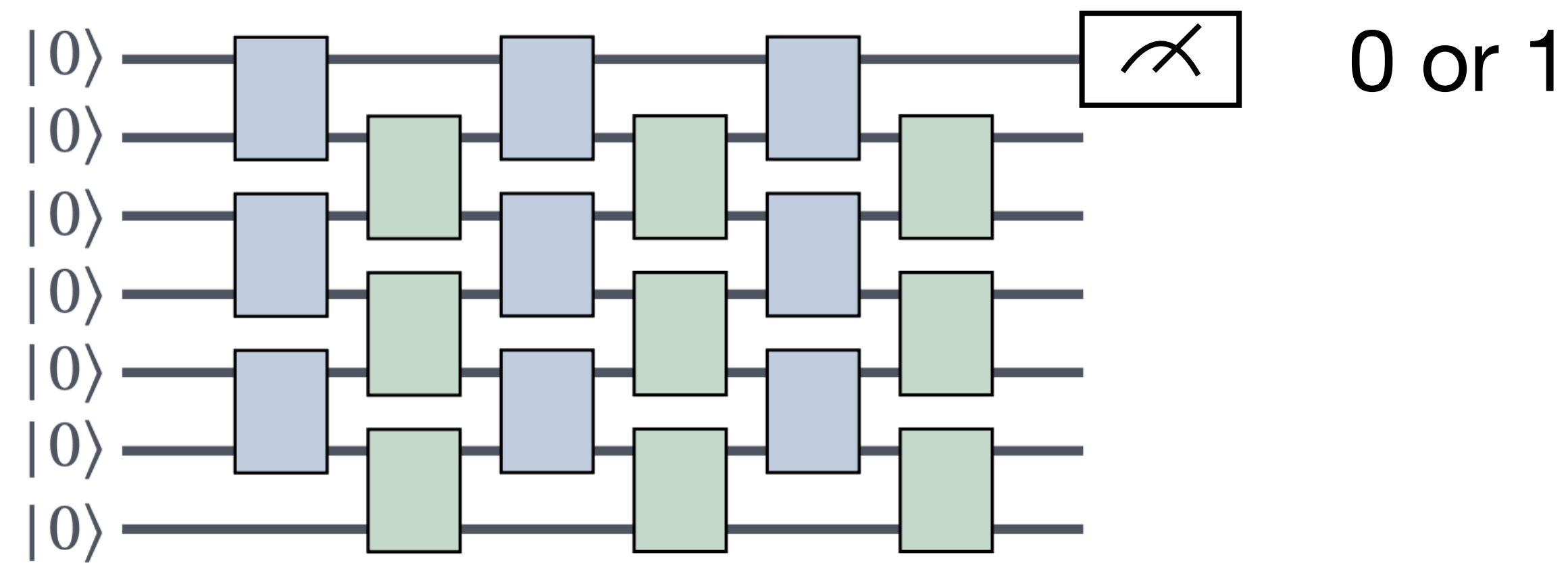
In practice, the goal is to generate a unitary transformation, i.e. a quantum circuit, such that $|\hat{U}_t - \hat{V}_t| < \epsilon$

In practice, time evolution is implemented approximately

Measurement

Quantum-mechanical measurement process is probabilistic.

Example: measurement in the Z basis



Measurement produces bit strings $\{x_k\}_k$ out of which expectation values can be computed, e.g.

$$\langle \Psi | Z | \Psi \rangle \simeq \mu \pm \frac{\sigma}{\sqrt{N-1}} \quad , \quad \mu = \frac{1}{N} \sum_k (1 - 2x_k) \quad , \quad \sigma^2 = \frac{1 - \mu^2}{N}$$

To resolve an expectation value within a target statistical uncertainty, one needs to accumulate enough measurement outcomes or “shots”.

RPMBT XXIII – Quantum Simulation School

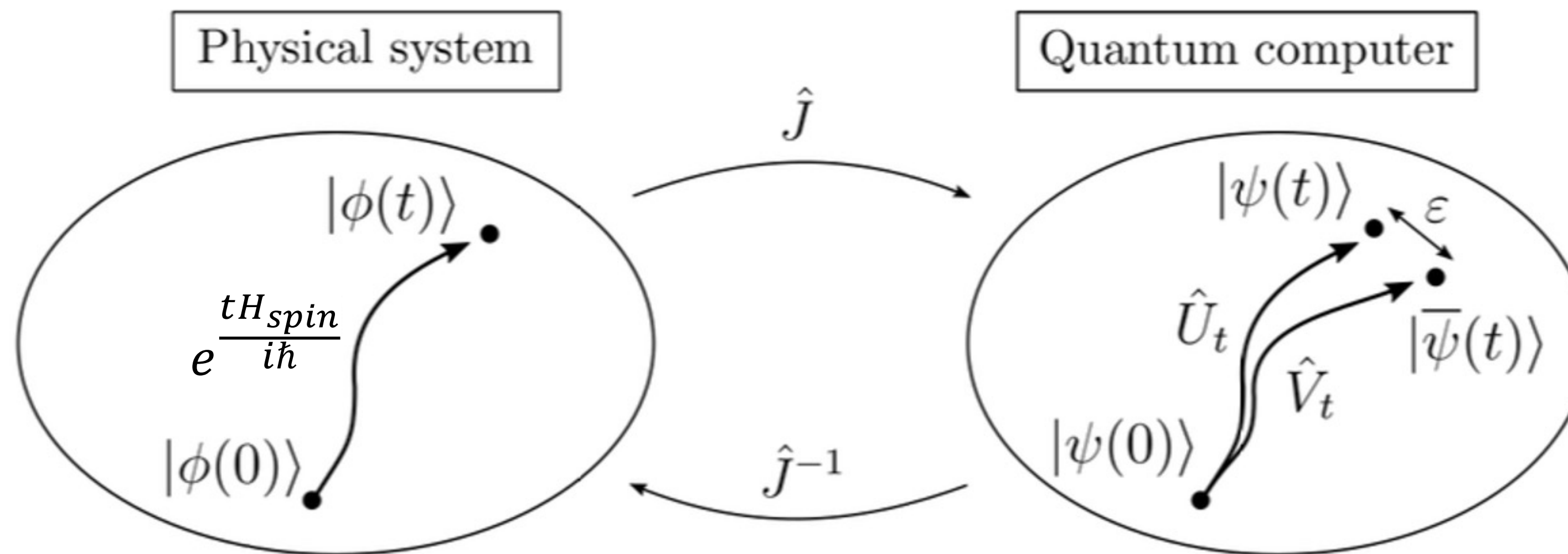
	9 Sept	10 Sept	11 Sept
8:30-9:00	Registration		
9:00-9:30	Welcome+Sponsors	THEORY 3 VQE, intro to errors and their mitigation	THEORY 5 ancilla-free spectral function, circuit knitting
9:30-10:30	THEORY 1 Quantum Simulations		
10:30-11:00	Break	Break	Break
11:00-12:30	EXERCISE 1 Time evolution for Ising chain	EXERCISE 3 VQE and read-out error mitigation	THEORY 6 advanced error mitigation, LCU and QSP
12:30-14:00	Lunch	Lunch	Lunch
14:00-14:20	THEORY 2 Trotter-Suzuki decomposition, linear response	LINKS Foundation	EXERCISE 5 full protocol implementation on hardware
14:20-15:30		THEORY 4 error mitigation for quantum gates	
15:30-16:00	Break	Break	Break
16:00-17:30	EXERCISE 2 Correlation functions and execution on emulators	EXERCISE 4 Execution on quantum hardware with full mitigation	EXERCISE 6 full protocol implementation on hardware, extensions

RPMBT XXIII – Quantum Simulation School

- Trotter approximation
- Trotter approximation for the Ising Hamiltonian
- Dynamical structure factor
- Hadamard test
- An application

Simulating dynamics with quantum circuits

Mapping states of n spins on states of n qubits defines an isomorphism J between the Hilbert spaces of spins and qubits. Operators are mapped as $B_{qubit} = JB_{spin}J^{-1}$



Under the isomorphism, the time-evolution operator takes the form $\hat{U}_t = e^{\frac{tH_{qubit}}{i\hbar}} = e^{\frac{tJH_{qubit}J^{-1}}{i\hbar}} = J e^{\frac{tH_{spin}}{i\hbar}} J^{-1}$

For generic Hamiltonians, the exact exponential maps to a circuit of exponential cost.

In practice, the goal is to generate a unitary transformation, i.e. a quantum circuit, such that $|\hat{U}_t - \hat{V}_t| < \epsilon$

In practice, time evolution is implemented approximately

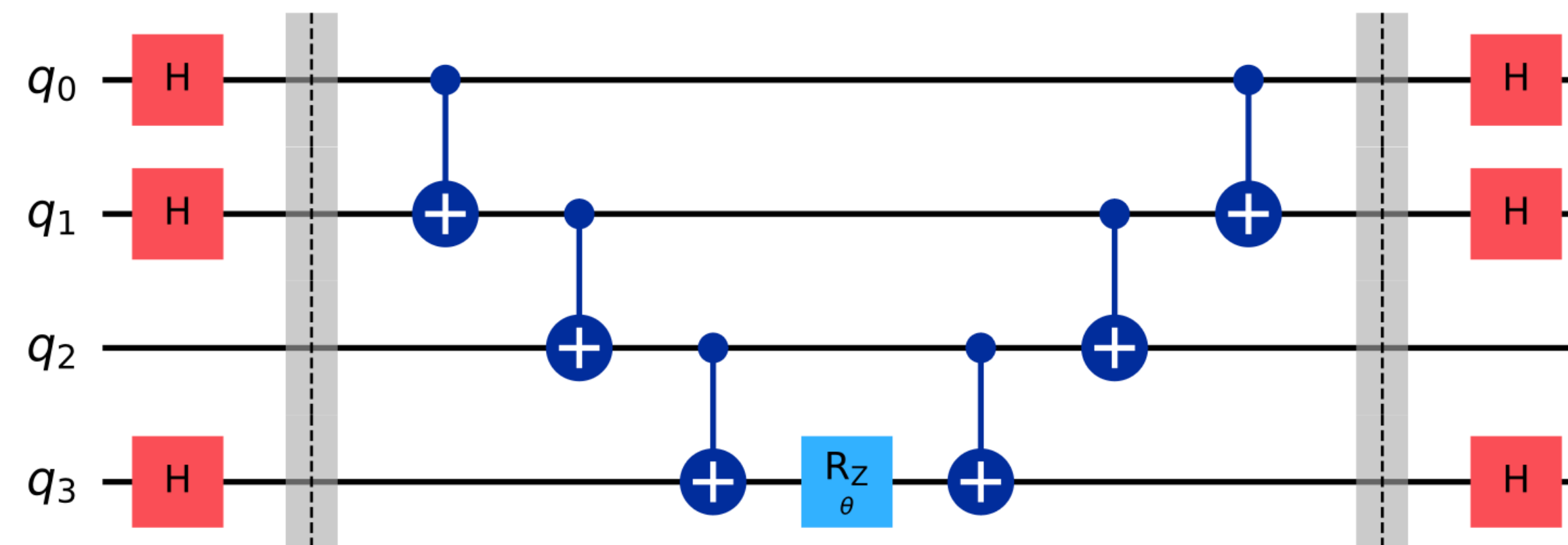
Time evolution under a single Pauli operator

For a single n-qubit Pauli operator, e.g. $H = \gamma XZXX = \gamma P$, the exponential can be computed efficiently and exactly.

1. Transform the Pauli operator into a tensor product of Z operators using $Z = \text{Had} X \text{Had}$, $Y = (\text{Had}S) Z (\text{Had}S)^\dagger$

$$P = X \otimes Z \otimes X \otimes X = (\text{Had} \otimes I \otimes \text{Had} \otimes \text{Had})(Z \otimes Z \otimes Z \otimes Z)(\text{Had} \otimes I \otimes \text{Had} \otimes \text{Had})$$

- 2) Implement $\exp(-it\gamma Z \dots Z)$ with $2(n - 1)$ CNOT gates and one Z-rotation by an angle $\theta = 2t\gamma$



Exact implementation, with $O(n)$ elementary gates for a generic time t

Product Formulas

Typical Hamiltonians are not single Pauli operators, but linear combinations of many Pauli operators

$$H = -J \sum_q X_q X_{q+1} - h \sum_q Z_q$$

If a Hamiltonian has the form $H = \sum_{m=1}^L c_m P_m$ and these Pauli operators commute with each other, then

$$e^{-itH} = \prod_{m=1}^L e^{-itc_m P_m}$$

Time evolution can then be simulated exactly for any time t with $O(nL)$ gates.
But what if the Pauli operators do not commute?

Product formulas = class of formulas that approximate e^{-itH} using products of the operators $e^{-i\gamma H_m}$

1. The first ingredient of product formulas is to break the time $[0, t]$ into n_s steps of duration $\Delta t = t/n_s$
2. The time-evolution operator for a single step is approximated with an error scaling as Δt^k for some $k \geq 2$
3. The error of product formulas decreases to zero when the number of step increases

Product formulas – primitive approximation

The “primitive approximation” is simply

$$U_{\Delta t} = e^{-i\Delta t H} \simeq e^{-i\Delta t H_L} \dots e^{-i\Delta t H_1} = V_{\Delta t}$$

The two operators differ at second order in Δt

$$U_{\Delta t} = e^{-i\Delta t H} = 1 + (-i\Delta t H) + (-i)^2 \Delta t^2 \sum_{ij} H_i H_j + \dots$$

$$V_{\Delta t} = e^{-i\Delta t H_L} \dots e^{-i\Delta t H_1} = 1 + (-i\Delta t H) + \frac{(-i\Delta t)^2}{2} \sum_i H_i^2 + \frac{(-i\Delta t)^2}{2} \sum_{i < j} 2H_i H_j + \dots$$

$$|U_{\Delta t} - V_{\Delta t}| = \left| \frac{(-i\Delta t)^2}{2} \sum_{i < j} [H_i, H_j] \right| + \dots \simeq C \Delta t^2$$

Simulation for time t with accuracy ε requires $n_s = O(Ct^2/\varepsilon)$ steps.

Simulation is efficient provided L, C grow polynomially with system size.

Product formulas — second-order Suzuki formula

The “2nd order Suzuki approximation” is

$$U_{\Delta t} = e^{-i\Delta t H} \simeq e^{-i\frac{\Delta t H_1}{2}} \dots e^{-i\frac{\Delta t H_L}{2}} e^{-i\frac{\Delta t H_L}{2}} \dots e^{-i\frac{\Delta t H_1}{2}} = V_{\Delta t}$$

The two operators differ at 3rd order in the time step

$$|U_{\Delta t} - V_{\Delta t}| = \tilde{C}\Delta t^3 + \dots$$

Primitive:

$$e^{-i\Delta t \hat{H}_1}$$

$$e^{-i\Delta t \hat{H}_2}$$

$$e^{-i\Delta t \hat{H}_3}$$

$$e^{-i\Delta t \hat{H}_4}$$

$$e^{-i\Delta t \hat{H}_5}$$

$$e^{-i\Delta t \hat{H}_6}$$

Suzuki:

$$e^{-\frac{i\Delta t \hat{H}_1}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_2}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_3}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_4}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_5}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_6}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_6}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_5}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_4}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_3}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_2}{2}}$$

$$e^{-\frac{i\Delta t \hat{H}_1}{2}}$$

Simulation for time t with accuracy ε requires $n_s = O(\tilde{C}^{1/2} t^{3/2} / \varepsilon^{1/2})$ steps.

Simulation is efficient provided $L, \tilde{C}$ grow polynomially with system size.

Trotter-Suzuki decomposition for the Ising model

- Hamiltonian

$$H = -J \sum_{\langle qr \rangle} X_q X_r - h \sum_q Z_q$$

- Decomposition in steps

$$e^{-in_s \Delta t H} = \left(e^{-i \Delta t H} \right)^{n_s}$$

- Suzuki approximation in a single step

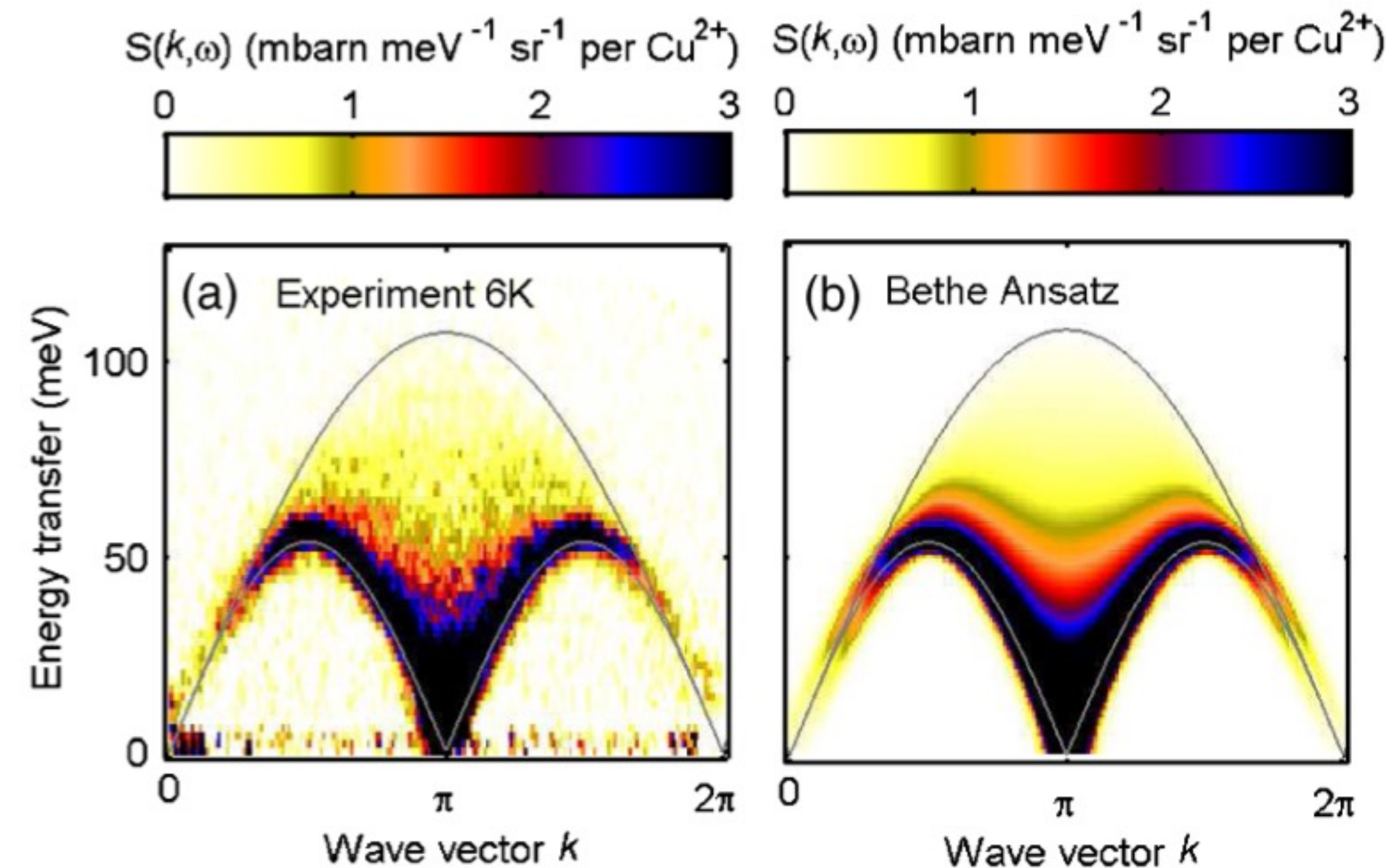
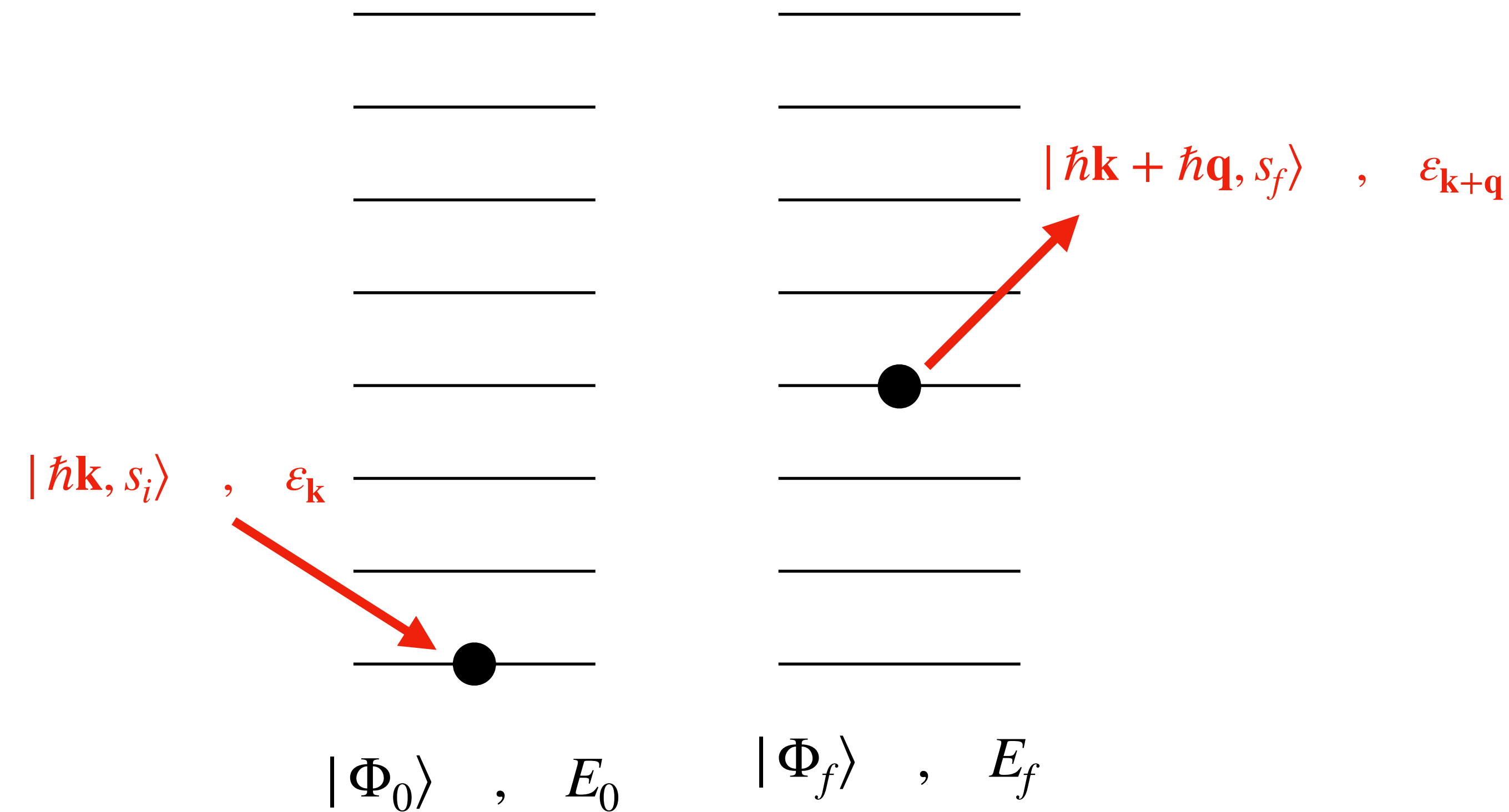
$$e^{-i \Delta t H} \simeq \left[e^{i \frac{\Delta t h}{2} \sum_q Z_q} \right] \left[e^{i \Delta t J \sum_{\langle qr \rangle} X_q X_r} \right] \left[e^{i \frac{\Delta t h}{2} \sum_q Z_q} \right]$$

- Representation as Pauli rotations $P(\varphi) = e^{-i \frac{\varphi}{2} P}$

$$e^{-i \Delta t H} \simeq \left[\prod_q Z_q(-\Delta t h) \right] \left[\prod_{\langle qr \rangle} X X_{qr}(-2 \Delta t J) \right] \left[\prod_q Z_q(-\Delta t h) \right]$$

Neutron scattering

1. A material at equilibrium in the ground state $|\Phi_0\rangle$ is hit with neutrons having momentum and spin $|\hbar\mathbf{k}, s_i\rangle$
2. The system transitions to an excited state $|\Phi_f\rangle$ and neutrons to $|\hbar\mathbf{k} + \hbar\mathbf{q}, s_f\rangle$
3. The intensity of neutrons is measured versus exchanged momentum $\hbar\mathbf{q}$ and energy $\hbar\omega = E_f - E_0 = \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}+\mathbf{q}}$

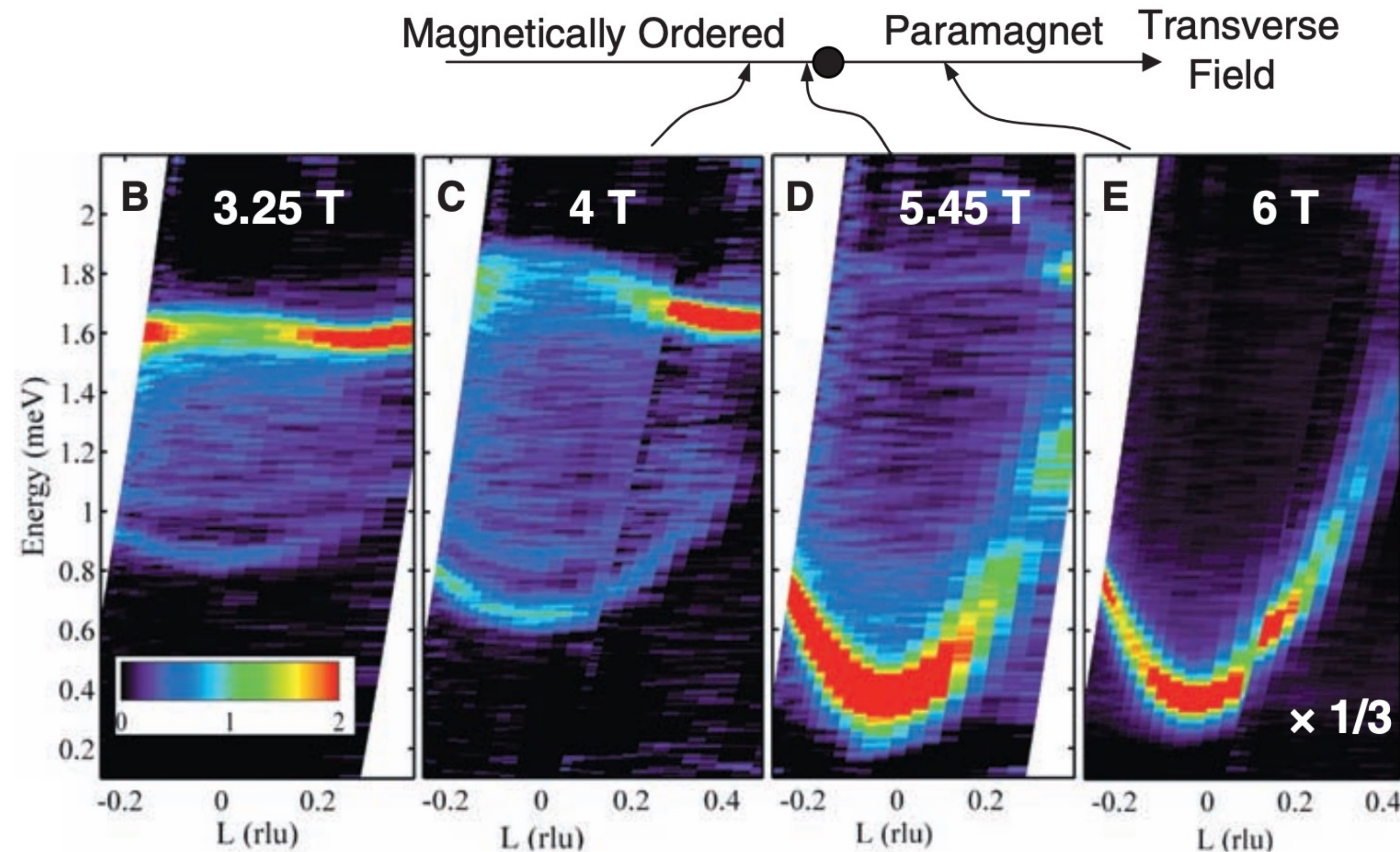


Neutron scattering from CoNb_2O_6

Dynamical structure factor as a function of external field h

Sharp transition at $h \simeq 5.45 \text{ T}$ from magnetically ordered to paramagnetic system

spin-flip cost $4J$



spin wave cost $2h - 2J \cos(k)$

Dynamical structure factor

Magnetic interaction between incident neutrons and spins in the system

$$V = \sum_{\ell} V(\mathbf{R} - \mathbf{R}_{\ell}) \mathbf{S} \cdot \mathbf{S}_{\ell} = \sum_{\mathbf{q}} V_{\mathbf{q}} [e^{i\mathbf{R} \cdot \mathbf{q}} \mathbf{S}] \cdot \left[\sum_{\ell} e^{-i\mathbf{R}_{\ell} \cdot \mathbf{q}} \mathbf{S}_{\ell} \right] = \sum_{\mathbf{q}} V_{\mathbf{q}} [e^{i\mathbf{R} \cdot \mathbf{q}} \mathbf{S}] \cdot \mathbf{B}_{\mathbf{q}}^{\dagger}$$

Interaction causes transitions in the state of neutrons and spins, at a rate given by **Fermi's golden rule**

$$R(\hbar\mathbf{k}, s_i, \Phi_0 \rightarrow \hbar\mathbf{k} + \hbar\mathbf{q}, s_f, \Phi_f) = \frac{2\pi}{\hbar} \delta(E_f - E_0 - \hbar\omega) |\langle \hbar\mathbf{k}, s_i, \Phi_0 | V | \hbar\mathbf{k} + \hbar\mathbf{q}, s_f, \Phi_f \rangle|^2$$

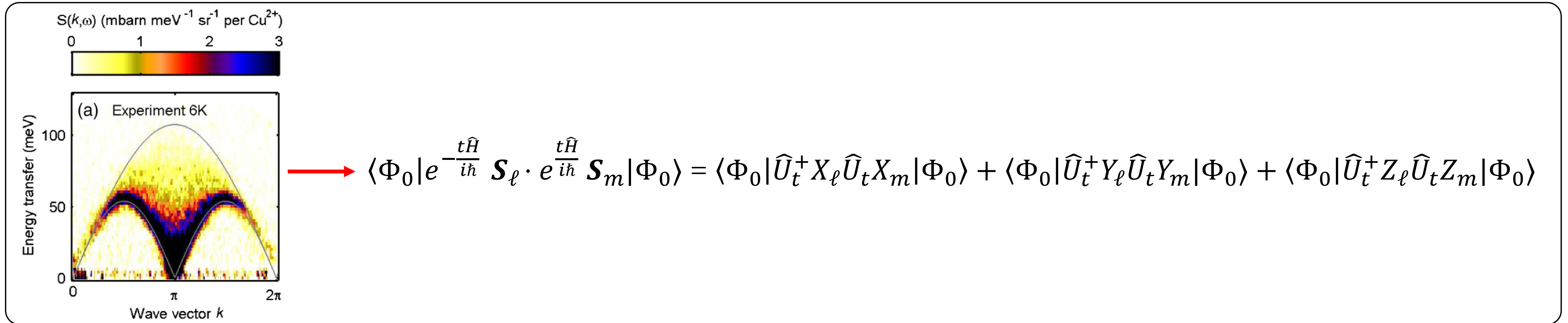
Intensity of neutrons is proportional to the **dynamical structure factor**

$$S(\mathbf{q}, \omega) = \sum_{f, s_f} R(\hbar\mathbf{k}, s_i, \Phi_0 \rightarrow \hbar\mathbf{k} + \hbar\mathbf{q}, s_f, \Phi_f) = |V_{\mathbf{q}}|^2 \langle \Phi_0 | \mathbf{B}_{\mathbf{q}}^{\dagger} \left[\sum_f \frac{2\pi}{\hbar} \delta(E_f - E_0 - \hbar\omega) | \Phi_f \rangle \langle \Phi_f | \right] \cdot \mathbf{B}_{\mathbf{q}} | \Phi_0 \rangle$$

The dynamical structure factor is the time-frequency and position-momentum Fourier transform of the **spin-spin correlation function**

$$S(\mathbf{q}, \omega) = \frac{|V_{\mathbf{q}}|^2}{\hbar^2} \sum_{\ell m} e^{i(\mathbf{R}_{\ell} - \mathbf{R}_m) \cdot \mathbf{q}} \int dt e^{i\omega t} \langle \Phi_0 | e^{-\frac{tH}{i\hbar}} \mathbf{S}_{\ell} \cdot e^{\frac{tH}{i\hbar}} \mathbf{S}_m | \Phi_0 \rangle$$

Measurement of time-dependent correlation functions



Computing response functions requires extra ingredients:

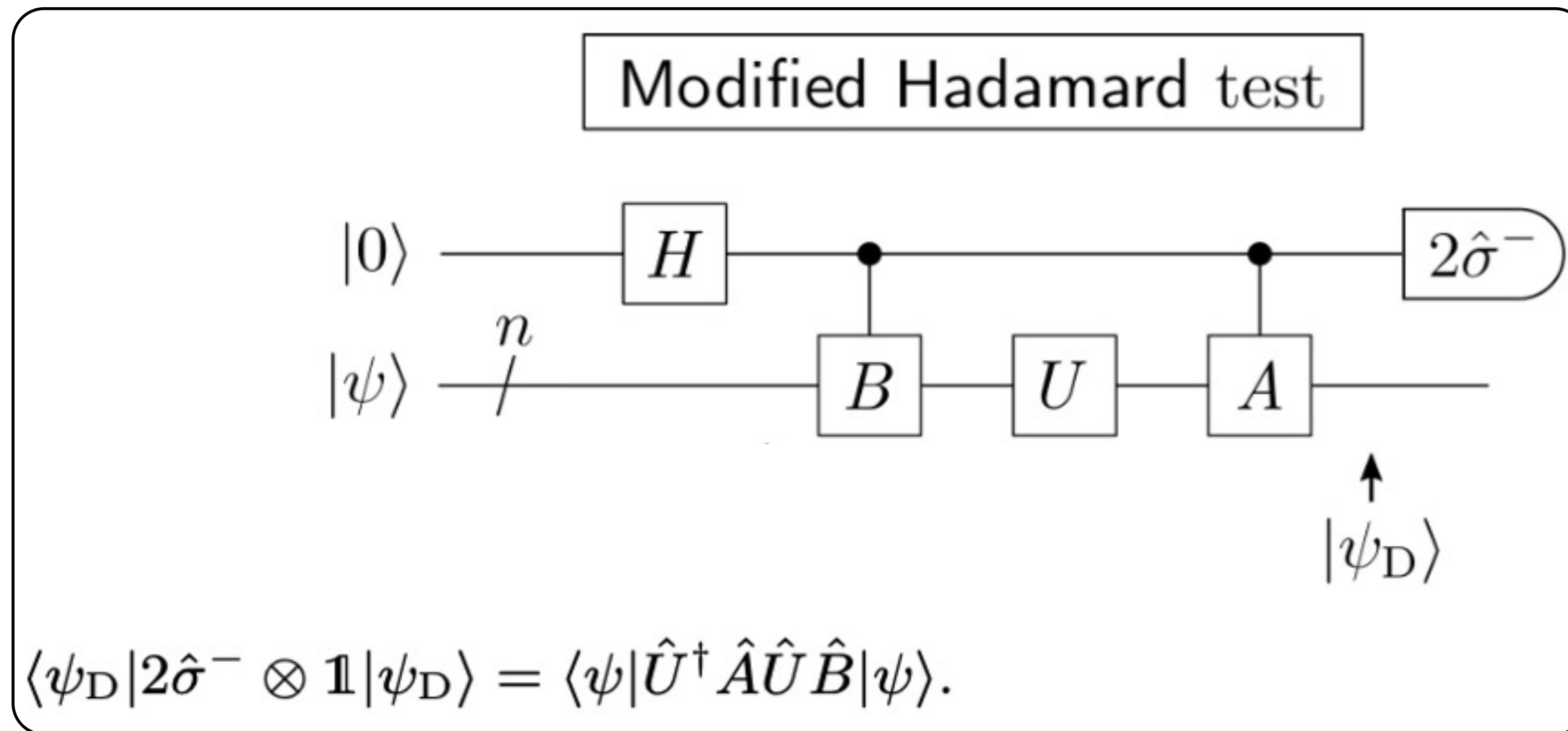
1. An approximation for the ground state [DAY 2]
2. A time evolution unitary [e.g. Trotter approximations seen before]
3. A measurement of a product of Pauli operators interspersed with time-evolution operators

This measurement is not simply that of a Hermitian operator!

How to perform this measurement?

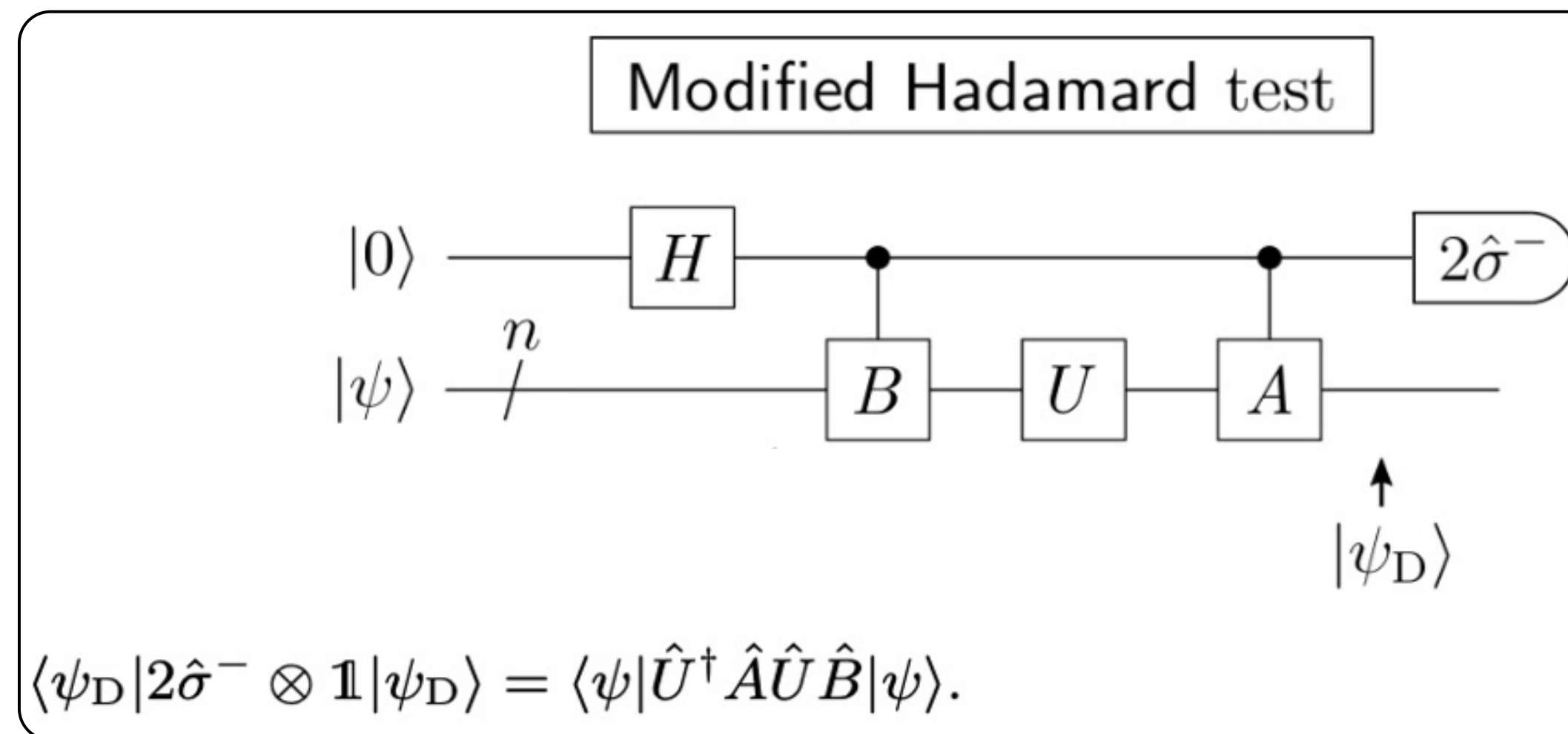
(Modified) Hadamard test

1. Append an extra qubit (ancillary) to the main register
2. Prepare the ancilla in a superposition of $|0\rangle$ and $|1\rangle$
3. Apply controlled Pauli operators A, B and the uncontrolled time evolution U
4. Measure the ancilla in the X and Y basis



Let us verify that this circuit perform the intended task...

(Modified) Hadamard test



Initial state $|0\rangle |\psi\rangle$

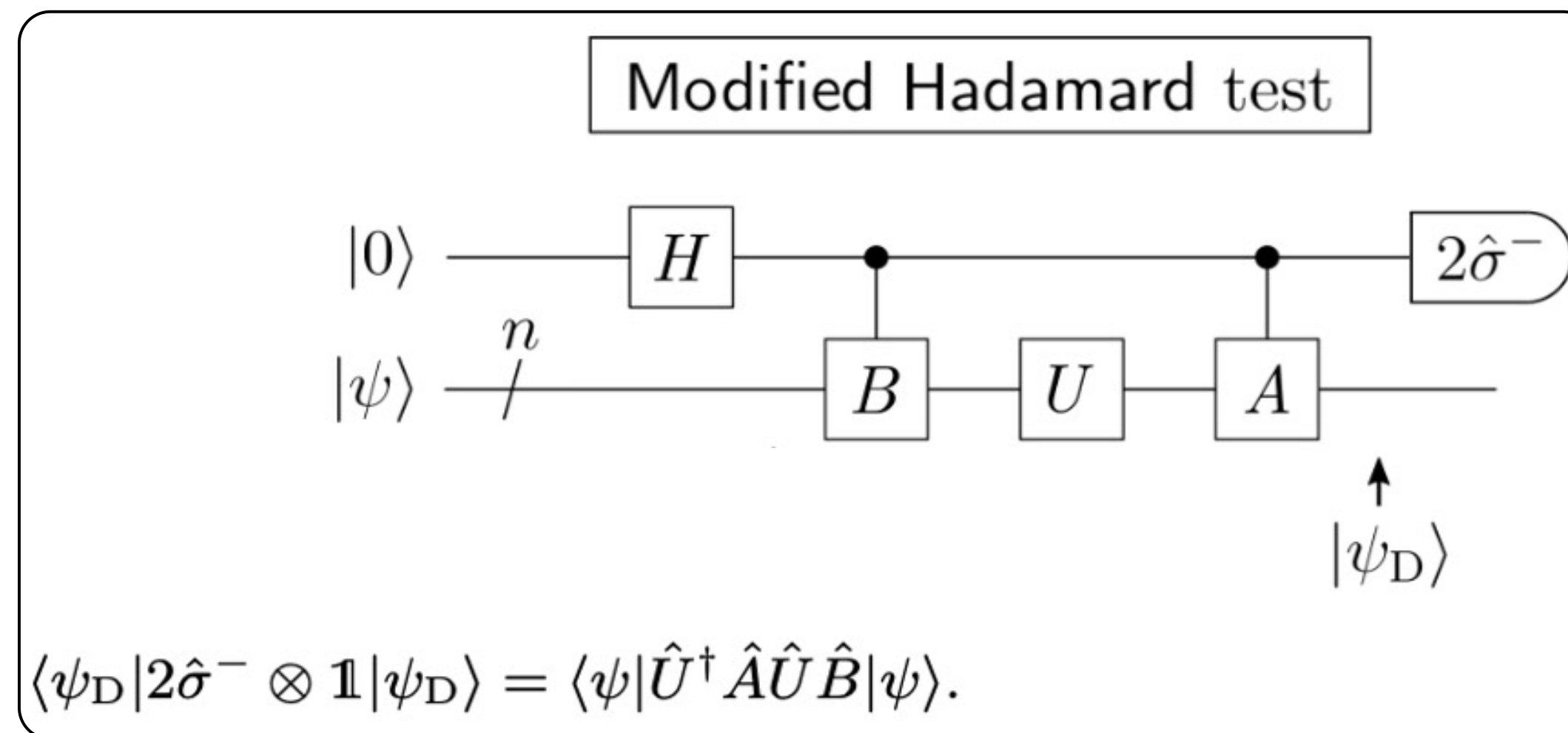
Hadamard gate $\frac{|0\rangle |\psi\rangle + |1\rangle |\psi\rangle}{\sqrt{2}}$

Apply B if the ancilla is in $|1\rangle$ to obtain $\frac{|0\rangle |\psi\rangle + |1\rangle B |\psi\rangle}{\sqrt{2}}$

Apply time evolution, uncontrolled $\frac{|0\rangle U |\psi\rangle + |1\rangle UB |\psi\rangle}{\sqrt{2}}$

Apply A if the ancilla is in $|1\rangle$ to obtain $\frac{|0\rangle U |\psi\rangle + |1\rangle AUB |\psi\rangle}{\sqrt{2}}$

(Modified) Hadamard test



Final state $\frac{|0\rangle U|\psi\rangle + |1\rangle AUB|\psi\rangle}{\sqrt{2}}$

If we measure the operator $2|0\rangle\langle 1| = (X + iY)$ over the ancilla, what do we get?

$$\frac{\langle 0 | \langle \psi | U^\dagger + \langle 1 | \langle \psi | (AUB)^\dagger}{\sqrt{2}} [2|0\rangle\langle 1| \otimes I] \frac{|0\rangle U|\psi\rangle + |1\rangle AUB|\psi\rangle}{\sqrt{2}}$$

$$\frac{\langle 0 | \langle \psi | U^\dagger}{\sqrt{2}} [2|0\rangle\langle 1| \otimes I] \frac{|1\rangle AUB|\psi\rangle}{\sqrt{2}} = \langle \psi | U^\dagger AUB | \psi \rangle$$

Dynamical structure factor — a recent application

Comparison between

- experiment
- quantum computation
- classical simulation

